

# A linear bound for the Paulsen problem

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## Abstract

We prove that every real  $\varepsilon$ -nearly equal-norm Parseval frame of  $n$  vectors in  $\mathbb{R}^d$  is within squared Frobenius distance  $C\varepsilon d$  of an equal-norm Parseval frame, where  $C$  is an absolute constant; this is optimal up to the value of  $C$ . Equivalently, every rank- $d$  orthogonal projection on  $\mathbb{R}^n$  whose diagonal entries deviate from  $d/n$  by at most  $\beta$  is within squared Frobenius distance  $Cn\beta$  of a rank- $d$  orthogonal projection with constant diagonal. The proof has two parts. A deterministic *static balancing theorem*, proved by minimising a log-determinant potential over a box, provides a row scaling that corrects every diagonal error that is small compared with the inverse of the *barrier constant*: a half-set hitting time of the random walk whose jump rates are the squared entries  $P_{ij}^2$  of the projection. Two random *seeds* then make the barrier constant small at linear cost: a filtered Gaussian tangent perturbation with a deterministic drift when  $d \gtrsim \log n$ , and a row-wise tangent perturbation with row renormalisation when  $n \gtrsim d^2$ . Bounded  $d$  is handled by the Hamilton–Moitra argument. The main theorem has been formalised in Lean 4 with Mathlib.

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# 1 The result and the plan

## 1.1 Statement

A frame of  $n$  vectors in  $\mathbb{R}^d$  is a matrix  $X \in \mathbb{R}^{n \times d}$  whose  $i$ th row is  $x_i^T$ . Distances are *squared* Frobenius distances  $\|X - Y\|_F^2 = \sum_i \|x_i - y_i\|^2$ , with no normalisation. Throughout,

$$a = d/n.$$

$X$  is *Parseval* if  $X^T X = I_d$  and *equal-norm* if  $\|x_i\|^2 = a$  for all  $i$ ; an equal-norm Parseval frame is an *ENP frame*.  $X$  is an  $\varepsilon$ -*nearly ENP frame* if

$$(1 - \varepsilon)I_d \preceq X^T X \preceq (1 + \varepsilon)I_d, \quad (1 - \varepsilon)a \leq \|x_i\|^2 \leq (1 + \varepsilon)a \quad (1 \leq i \leq n). \quad (1.1)$$

**Theorem 1.1** (Linear Paulsen bound). *There is an absolute constant  $C$  such that for all  $1 \leq d \leq n$ , all  $0 < \varepsilon < 1$  and every real  $\varepsilon$ -nearly ENP frame  $X \in \mathbb{R}^{n \times d}$  there is an ENP frame  $W \in \mathbb{R}^{n \times d}$  with  $\|X - W\|_F^2 \leq C\varepsilon d$ .*

The bound is optimal up to the constant (Remark 1.3). Kwok, Lau, Lee and Ramachandran [3] proved the first bound independent of  $n$ ,  $O(\varepsilon d^{13/2})$ ; Hamilton and Moitra [2] improved it to  $O(\varepsilon d^2)$ ; Lau and Ramachandran announced the linear bound [4], via a different route. Appendix B describes a Lean 4 formalisation of Theorems 1.1 and 1.2.

For a Parseval  $U$  write  $P = UU^T$  for the orthogonal projection onto its column space; then  $P_{ii} = \|u_i\|^2$ . The problem has an equivalent form in which the error is additive.

**Theorem 1.2** (Projection form). *There is an absolute constant  $C$  such that for every  $n \geq 1$  and every rank- $d$  orthogonal projection  $P$  on  $\mathbb{R}^n$ , with  $\beta = \max_i |P_{ii} - d/n|$ , there is a rank- $d$  orthogonal projection  $Q$  with  $Q_{ii} = d/n$  for all  $i$  and  $\|P - Q\|_F^2 \leq Cn\beta$ .*

This reformulation and the complement symmetry are due to Cahill and Casazza [1]: both  $\beta$  and  $\|P - Q\|_F$  are unchanged when  $(P, Q)$  is replaced by  $(I - P, I - Q)$ , so in Theorem 1.2 one may assume  $d \leq n/2$ . In Section 6, Theorem 1.2 is derived from an equal-norm formulation (Proposition 6.1) and Theorem 1.1 from Theorem 1.2.

*Remark 1.3 (Optimality).* Let  $d = 2d_1$ ,  $n = 2n_1$ , and  $1 \leq k \leq n_1/4$  with  $n_1 - k \geq d_1$ ; put  $\varepsilon = 4k/n$ . Let  $X = X_1 \oplus X_2$ , where  $X_1$  is an ENP frame of  $n_1 - k$  vectors in  $\mathbb{R}^{d_1}$  and  $X_2$  one of  $n_1 + k$  vectors in  $\mathbb{R}^{d_1}$  (Corollary 3.2). Then  $X$  is Parseval, and its squared row norms  $a(1 \mp 2k/n)^{-1}$  make it an  $\varepsilon$ -nearly ENP frame. Let  $W$  be ENP, and let  $E$  and  $A$  be the principal blocks of  $XX^T$  and  $WW^T$  on the first  $n_1 - k$  rows. Then  $E$  is a projection of trace  $d_1$  and  $A$  a positive contraction of trace  $(n_1 - k)a$ , so  $\Delta = E - A$  has  $\text{tr } \Delta = ka = \varepsilon d/4$ . Since  $\text{tr}(E\Delta) = \text{tr } \Delta + \text{tr}((I - E)A) \geq \text{tr } \Delta$ ,

$$\text{tr}(A - A^2) = -\text{tr } \Delta + 2\text{tr}(E\Delta) - \text{tr } \Delta^2 \geq \text{tr } \Delta - \|\Delta\|_F^2.$$

As  $WW^T$  is a projection, its off-diagonal block has squared norm  $\text{tr}(A - A^2)$ , while that of  $XX^T$  vanishes; hence  $\|XX^T - WW^T\|_F^2 \geq \|\Delta\|_F^2 + 2(\text{tr } \Delta - \|\Delta\|_F^2) \geq 2\text{tr } \Delta - \|XX^T - WW^T\|_F^2$ , i.e.  $\|XX^T - WW^T\|_F^2 \geq \text{tr } \Delta$ , and by Lemma 2.1  $\|X - W\|_F^2 \geq \frac{1}{2}\text{tr } \Delta = \varepsilon d/8$ .

**Conventions.**  $c, C$  denote positive absolute constants whose value may change between occurrences; constants with subscripts or names are fixed once chosen.  $\|\cdot\|_{\text{op}}$  is the Euclidean operator norm,  $\mathbf{1}$  the all-ones vector,  $\text{Diag}(x)$  the diagonal matrix with diagonal  $x$ ,  $\text{diag}(M)$  the diagonal of  $M$  as a vector, and  $\Pi_{\mathcal{S}}$  the orthogonal projection onto a subspace  $\mathcal{S}$ .  $A \preceq B$  is the Loewner order. All Gaussian vectors are real and centred.

## 1.2 The plan

**The engine: scaling.** Let  $U$  be Parseval with projection  $P$  and diagonal  $p = \text{diag } P$ . For  $s \in \mathbb{R}^n$  the matrix  $\text{Diag}(e^s)U$  has a column space with projection  $P(s)$ , and  $s \mapsto \text{diag } P(s)$  is the gradient of the convex function  $\frac{1}{2} \log \det(U^T \text{Diag}(e^{2s})U)$ , whose Hessian at  $s = 0$  is  $2L_P$ , where

$$L_P = \text{Diag}(p) - P \circ P$$

is the graph Laplacian with edge weights  $P_{ij}^2$  (Lemma 2.2). To first order, changing the diagonal by  $\delta \perp \mathbf{1}$  therefore requires  $s \approx \frac{1}{2}L_P^+ \delta$ , at a cost of about  $s^T L_P s \approx \frac{1}{4} \delta^T L_P^+ \delta$ . If  $L_P$  has spectral gap  $\lambda$  and  $|\delta_i| \leq \varepsilon a$ , the cost is at most  $n\varepsilon^2 a^2/\lambda$ . For a “random-looking”  $P$  one has  $\lambda \approx a$  and the cost is  $\varepsilon^2 d$ , but  $L_P$  can be arbitrarily badly connected: for an exact direct sum the components cannot exchange diagonal mass at all.

**Perturb, then scale.** Perturb  $U$  by a random horizontal tangent of size  $t = \sqrt{K_0 \varepsilon}$ , with  $K_0$  a large constant, at cost  $O(t^2 d) = O(\varepsilon d)$ . Generic entries of the new projection then have size  $t\sqrt{a/n}$ , so most edges acquire weight  $\gtrsim at^2/n$ , and the random walk with rates  $P_{ij}^2$  reaches any half of the vertices in time  $O(1/(at^2))$ : the *barrier constant* (Definition 2.6 and Lemma 2.10) is  $O(1/(at^2))$ . The static balancing theorem (Theorem 2.8) then removes exactly every diagonal error  $\beta$  with  $\beta \cdot O(1/(at^2)) \leq \frac{1}{2}$ , at cost  $\frac{15}{4}n\beta = O(t^2 d)$ . The  $\ell_\infty$  (rather than spectral) formulation is essential: the scaling must stay in a box  $\|s\|_\infty \leq 1$ , and an  $\ell_2$  argument loses a factor  $\sqrt{n}$ .

**The difficulty.** The perturbed frame must therefore have diagonal error at most  $\theta at^2$  in  $\ell_\infty$ , for a small fixed  $\theta$ . The initial error  $\varepsilon a = at^2/K_0$  is acceptable. But a random tangent of size  $t$  moves each diagonal entry *linearly* by about  $t\sqrt{a/n}$ , which exceeds  $at^2$  as soon as  $\varepsilon \lesssim 1/d$ . Two different constructions deal with this.

**Seed 1: many rows** ( $n \geq Bd^2$ , [Section 4](#)). Here the input is an equal-norm frame  $X$  with spectral error  $\eta = \|X^T X - I\|_{\text{op}}$ , and  $t^2$  is a large multiple of  $\eta$ . Perturb each row *orthogonally to itself* and renormalise. Row norms stay exact, but the frame operator moves. Its first-order change is removed by conditioning the Gaussian on a subspace of codimension at most  $d^2$ , and its second-order fluctuation is an average over  $n$  rows, of size  $t^2 d / \sqrt{n}$ . The spectral error of the perturbed frame is therefore  $\eta + O(t^2 d / \sqrt{n} + t^2 d^2 / n + t^3)$ , which is at most  $\theta t^2$  once  $t^2 / \eta$  and  $B$  are large and  $t$  is small; whitening turns a spectral error  $\delta$  into a diagonal error at most  $2a\delta$ . The cubic term is  $O(t^3)$ , not the termwise  $O(t^3 d)$ , because the first-order constraint allows the normalisation weights to be centred ([Lemma 4.4](#)); this is why the seed works for every  $\eta \leq c_*$ , with no relation required between  $\eta$  and  $d$ .

**Seed 2: moderate rows** ( $d \geq A \log 2n$ , [Section 5](#)). Perturb  $U$  in the horizontal tangent space of the Grassmannian, so that the frame stays exactly Parseval, with a Gaussian whose covariance is filtered:  $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$ , where  $\Omega$  is the normalised Gram operator of the linear diagonal map  $\mathcal{A}$  and  $\rho = D^2 t^2$  for a large constant  $D$ . The filter makes the linear diagonal change  $O(t\sqrt{\rho a \log n / n}) = O(Dat^2 \sqrt{\log n / d})$ , negligible when  $d \gg \log n$ , while leaving the weak (connecting) directions untouched. The price is a second-order bias  $t^2 b_*$ , which is not small in  $\ell_\infty$ . A commutator identity shows that  $b_* = \mathcal{A}m_*$  for an explicit deterministic tangent direction  $m_*$  with  $\|m_*\|_{\text{F}}^2 \leq 2a/\rho$  ([Lemmas 5.5](#) and [5.6](#)). We add the drift  $\frac{t}{2}m_*$  to the noise: its first-order effect cancels the bias exactly, and its other effects are  $O(at^2/D)$ . All remaining errors are at most  $\theta at^2$  by concentration, which is where  $d \geq A \log 2n$  is used. Only a bounded, deterministic set of rows needs a spectral argument, and second moments suffice there.

**Assembly** ([Section 6](#)). After complementation and normalisation it suffices to treat equal-norm frames with  $2d \leq n$  and spectral error  $\eta$ . If  $d < d_0$  (a large absolute constant), the Hamilton–Moitra bound  $\eta d(d-1) \leq d_0 \eta d$  suffices ([Section 3](#)). If  $d \geq d_0$  and  $n \geq Bd^2$ , Seed 1 applies. If  $d \geq d_0$  and  $n < Bd^2$ , then  $A \log 2n \leq A \log(2Bd^2) \leq d$  and Seed 2 applies.

| regime                         | construction                                                     | where                     |
|--------------------------------|------------------------------------------------------------------|---------------------------|
| $d < d_0$                      | radial isotropic position, ordered coordinates (Hamilton–Moitra) | <a href="#">Section 3</a> |
| $d \geq d_0, n \geq Bd^2$      | row-tangent Gaussian, row normalisation, whitening, balancing    | <a href="#">Section 4</a> |
| $d \geq d_0, 2d \leq n < Bd^2$ | filtered horizontal Gaussian with drift, balancing               | <a href="#">Section 5</a> |
| $d > n/2$                      | complement                                                       | <a href="#">Section 6</a> |

## 2 The scaling toolbox

Throughout this section  $U \in \mathbb{R}^{n \times d}$  is Parseval,  $P = UU^T$ ,  $p = \text{diag } P$ , and  $L_P = \text{Diag}(p) - P \circ P$ . For nonnegative symmetric weights  $w_{ij}$  on  $[n]$  the weighted Laplacian  $L$  acts by  $(Lx)_i = \sum_j w_{ij}(x_i - x_j)$ ; diagonal weights do not contribute. Unless another Laplacian is specified,  $L = L_P$ . We write  $\text{osc}(x) = \max_i x_i - \min_i x_i$ .

### 2.1 Projections, distances, and the squared-Gram Laplacian

**Lemma 2.1** (Frames versus projections). *Let  $U, W \in \mathbb{R}^{n \times d}$  be Parseval with projections  $P, Q$ , and let  $\sigma_1, \dots, \sigma_d \in [0, 1]$  be the singular values of  $U^T W$ . Then*

$$\min_{R \in O(d)} \|U - WR\|_{\text{F}}^2 = 2 \sum_k (1 - \sigma_k) \leq \|P - Q\|_{\text{F}}^2 = 2 \sum_k (1 - \sigma_k^2) = 2\|(I - P)W\|_{\text{F}}^2 \leq 2\|U - W\|_{\text{F}}^2.$$

Right multiplication by  $R \in O(d)$  preserves Parsevalness and every row norm. If  $X \in \mathbb{R}^{n \times d}$  has  $\delta = \|X^T X - I\|_{\text{op}} < 1$ , its polar factor  $\tilde{X} = X(X^T X)^{-1/2}$  is Parseval,  $\|X - \tilde{X}\|_{\text{F}}^2 \leq d\delta^2$ , and  $\|x_i\|^2/(1+\delta) \leq \|\tilde{x}_i\|^2 \leq \|x_i\|^2/(1-\delta)$ . If moreover  $\delta \leq \frac{1}{2}$ , then  $|\|\tilde{x}_i\|^2 - \|x_i\|^2| \leq 2\delta\|x_i\|^2$  and row  $i$  of  $\tilde{X}\tilde{X}^T - XX^T$  has squared norm at most  $6\delta^2\|x_i\|^2$ .

*Proof.*  $\|U - WR\|_{\text{F}}^2 = 2d - 2\text{tr}(U^T WR)$ , maximised over  $R$  by a polar factor of  $U^T W$  (any orthogonal  $R$  with  $U^T WR \succeq 0$ ; it need not be unique), giving  $2d - 2\sum_k \sigma_k$ . Next  $\|P - Q\|_{\text{F}}^2 = 2d - 2\text{tr}(PQ) = 2d - 2\|U^T W\|_{\text{F}}^2$  and  $\|(I - P)W\|_{\text{F}}^2 = d - \|U^T W\|_{\text{F}}^2$ ; also  $1 - \sigma \leq 1 - \sigma^2$  on  $[0, 1]$ , and  $(I - P)W = (I - P)(W - U)$  gives  $\|(I - P)W\|_{\text{F}} \leq \|U - W\|_{\text{F}}$ . For the polar factor, if  $\varsigma_k$  are the singular values of  $X$  then  $\|X - \tilde{X}\|_{\text{F}}^2 = \sum_k (\varsigma_k - 1)^2 \leq \sum_k (\varsigma_k^2 - 1)^2$ , and  $\|\tilde{x}_i\|^2 = x_i^T (X^T X)^{-1} x_i$ . With  $E = (X^T X)^{-1} - I$ , so  $\|E\|_{\text{op}} \leq \delta/(1 - \delta) \leq 2\delta$ , row  $i$  of  $\tilde{X}\tilde{X}^T - XX^T = XEX^T$  is  $XE x_i$ , of squared norm at most  $(1 + \delta)4\delta^2\|x_i\|^2$ .  $\square$

**Lemma 2.2** (Energy identity).  $L$  is the Laplacian of the weights  $P_{ij}^2$  ( $i \neq j$ ),  $L \succeq 0$ ,  $L\mathbf{1} = 0$ ,  $L_{I-P} = L_P$ , and for  $x \in \mathbb{R}^n$ ,  $X = \text{Diag}(x)$ ,

$$x^T Lx = \frac{1}{2} \sum_{i,j} P_{ij}^2 (x_i - x_j)^2 = \|(I - P)XU\|_{\text{F}}^2 = \frac{1}{2} \| [X, P] \|_{\text{F}}^2.$$

*Proof.*  $\sum_j P_{ij}^2 = (P^2)_{ii} = p_i$ , so the rows of  $\text{Diag}(p) - P \circ P$  sum to zero and its off-diagonal entries are  $-P_{ij}^2$ . Next  $\|(I - P)XU\|_{\text{F}}^2 = \text{tr}(XPX(I - P)) = \sum_i x_i^2 p_i - \sum_{i,j} x_i x_j P_{ij}^2$ , and  $[X, P]_{ij} = (x_i - x_j)P_{ij}$ . For  $I - P$  the off-diagonal weights are again  $P_{ij}^2$ .  $\square$

**Lemma 2.3** (Scaled projections). For  $w \in \mathbb{R}_{>0}^n$  put  $D = \text{Diag}(w)$ ,  $M = (U^T D^2 U)^{-1/2}$  and  $W = DUM$ . Then  $W$  is Parseval with column space  $D \text{im } U$ , and

$$\|WW^T - P\|_{\text{F}}^2 = 2\|(I - P)W\|_{\text{F}}^2 \leq \frac{2w^T Lw}{\min_i w_i^2}.$$

*Proof.*  $W^T W = MU^T D^2 U M = I$  and  $U^T D^2 U \succeq (\min_i w_i)^2 I$ , so  $\|M\|_{\text{op}} \leq 1/\min_i w_i$ . By Lemmas 2.1 and 2.2,  $\|(I - P)W\|_{\text{F}}^2 \leq \|M\|_{\text{op}}^2 \|(I - P)DU\|_{\text{F}}^2 = \|M\|_{\text{op}}^2 w^T Lw$ .  $\square$

## 2.2 Diagonal scaling

For  $s \in \mathbb{R}^n$  let  $D_s = \text{Diag}(e^s)$ , let  $P(s)$  be the projection onto  $D_s \text{im } U$ ,  $r(s) = \text{diag } P(s)$ , and  $F(s) = \frac{1}{2} \log \det(U^T D_s^2 U)$ .

**Lemma 2.4.**  $F$  is smooth and convex,  $\nabla F(s) = r(s)$ ,  $\sum_i r_i(s) = d$ , and  $F(s + c\mathbf{1}) = F(s) + cd$ .

*Proof.* With  $G = U^T D_s^2 U = \sum_i e^{2s_i} u_i u_i^T$  one has  $\partial_i F = e^{2s_i} u_i^T G^{-1} u_i = P(s)_{ii}$ , because  $P(s) = D_s U G^{-1} U^T D_s$ ; differentiating again gives  $\nabla^2 F = 2L_{P(s)} \succeq 0$ . The trace of  $P(s)$  is  $d$ , and  $G$  scales by  $e^{2c}$  under  $s \mapsto s + c\mathbf{1}$ .  $\square$

**Lemma 2.5** (One-sided scaling inequalities). For  $z \in \mathbb{R}_{>0}^n$  let  $r$  be the diagonal of the projection onto  $\text{Diag}(\sqrt{z}) \text{im } U$ . Then, coordinatewise,

$$Lz \leq z \circ (r - p), \quad L(z^{-1}) \leq -z^{-1} \circ (r - p).$$

*Proof.* Let  $G = U^T \text{Diag}(z)U$ . Since  $G + z_i^2 G^{-1} - 2z_i I = (G^{1/2} - z_i G^{-1/2})^2 \succeq 0$ , evaluating at  $u_i$  and using  $u_i^T G u_i = \sum_j P_{ij}^2 z_j = p_i z_i - (Lz)_i$  and  $z_i^2 u_i^T G^{-1} u_i = z_i r_i$  gives the first inequality. Let  $V \in \mathbb{R}^{n \times (n-d)}$  be an isometry onto  $(\text{im } U)^\perp$ . The subspaces  $\text{Diag}(\sqrt{z}) \text{im } U$  and  $\text{Diag}(z^{-1/2}) \text{im } V$  are orthogonal complements, so the projection onto the latter has diagonal  $\mathbf{1} - r$ , while  $VV^T = I - P$  has diagonal  $\mathbf{1} - p$  and Laplacian  $L$  (Lemma 2.2). The first inequality for  $(V, z^{-1})$  is the second. (If  $d = n$  then  $L = 0$  and  $r = p = \mathbf{1}$ .)  $\square$

### 2.3 The barrier constant and static balancing

**Definition 2.6** (Barrier constant). Let  $L$  be a weighted Laplacian on  $[n]$ . For  $S \subseteq [n]$ , an  $S$ -barrier is a vector  $\psi \geq 0$  with  $(L\psi)_i \geq 1$  for every  $i \notin S$ . The barrier constant  $H(L) \in (0, \infty]$  is the least  $H$  such that every  $S$  with  $|S| \geq n/2$  has an  $S$ -barrier with  $\|\psi\|_\infty \leq H$  ( $H(L) = \infty$  if some such  $S$  has no barrier).

For the continuous-time random walk that jumps from  $i$  to  $j$  at rate  $w_{ij}$ , the expected hitting time  $h_S(x) = \mathbb{E}_x \tau_S$  of  $S$  vanishes on  $S$  and satisfies  $(Lh_S)_i = 1$  off  $S$ . If  $\psi$  is any  $S$ -barrier and  $\beta' > 1$ , the maximum principle argument in the proof of Lemma 2.7, applied to  $h_S - \beta'\psi$ , gives  $h_S \leq \beta'\psi$ . So  $h_S$  is the minimal  $S$ -barrier, and  $H(L_P)$  is the half-set hitting time of the walk with rates  $P_{ij}^2$ : the largest expected time, over all starting points and all  $S$  with  $|S| \geq n/2$ , needed to reach  $S$ . If  $H(L) < \infty$  then the graph is connected: for a component  $J$  with  $|J| \leq n/2$ , summing  $(L\psi)_i$  over  $i \in J$  gives 0, so  $S = [n] \setminus J$  has no barrier. The barrier constant is within a factor 2 of the  $\ell_\infty$  Poisson constant of  $L$  (Remark 2.11). A small barrier constant controls scalings in  $\ell_\infty$ :

**Lemma 2.7** (Median barrier). Let  $L$  be a weighted Laplacian on  $[n]$  with  $H(L) \leq H < \infty$ , let  $\beta \geq 0$  with  $\beta H < 1$ , let  $z \in \mathbb{R}_{>0}^n$ , and let  $m$  be a median of  $z$  (so that  $\{z \leq m\}$  and  $\{z \geq m\}$  both have at least  $n/2$  elements). If

$$(Lz)_i \leq \beta z_i \text{ when } z_i > m, \quad (Lz^{-1})_i \leq \beta z_i^{-1} \text{ when } z_i < m,$$

then  $(1 - \beta H) \max_i z_i \leq m \leq (1 - \beta H)^{-1} \min_i z_i$ .

*Proof.* Let  $S = \{z \leq m\}$ , let  $\psi$  be an  $S$ -barrier with  $\|\psi\|_\infty \leq H$ , let  $M = \max z$  and  $\beta' > \beta$ . The vector  $v = z - m\mathbf{1} - \beta'M\psi$  is  $\leq 0$  on  $S$ , and for  $i \notin S$ ,  $(Lv)_i \leq \beta z_i - \beta'M < 0$ . At a maximiser  $i$  of  $v$ ,  $(Lv)_i = \sum_j w_{ij}(v_i - v_j) \geq 0$ ; so every maximiser lies in  $S$ , and  $v \leq 0$ . Hence  $M \leq m + \beta'MH$ ; let  $\beta' \downarrow \beta$ . Apply the same argument to  $z^{-1}$ , whose median is  $m^{-1}$ , with  $S = \{z \geq m\}$ .  $\square$

**Theorem 2.8** (Static balancing). Suppose  $H(L_P) \leq H < \infty$ , and  $q \in \mathbb{R}^n$  satisfies  $\sum_i q_i = d$  and  $\beta H \leq \frac{1}{2}$ , where  $\beta = \|q - p\|_\infty$ . Then there is a Parseval  $W$  with  $\text{diag}(WW^T) = q$  and

$$\|U - W\|_F^2 \leq \|UU^T - WW^T\|_F^2 \leq \frac{15}{4} \|q - p\|_1.$$

$W$  is obtained from  $U$  by a row scaling with ratio of scales at most 2, whitening, and an orthogonal right factor.

*Proof.* Let  $s$  minimise  $\Phi(s) = F(s) - \langle q, s \rangle$  over the cube  $[0, 1]^n$  (compactness), and write  $r = r(s)$ . Since  $\partial_i \Phi = r_i - q_i$  (Lemma 2.4), coordinatewise optimality gives

$$r_i \leq q_i \text{ if } s_i > 0, \quad r_i \geq q_i \text{ if } s_i < 1. \quad (2.1)$$

Let  $z = e^{2s}$  with median  $m$ , so  $1 \leq \min z \leq m \leq \max z \leq e^2$ . If  $z_i > m$  then  $s_i > 0$ , so by Lemma 2.5 and (2.1)  $(Lz)_i \leq z_i(r_i - p_i) \leq z_i(q_i - p_i) \leq \beta z_i$ ; if  $z_i < m$  then  $s_i < 1$  and likewise  $(Lz^{-1})_i \leq \beta z_i^{-1}$ . By Lemma 2.7,  $\max z / \min z \leq (1 - \beta H)^{-2} \leq 4$ , i.e.  $\text{osc}(s) \leq \log 2 < 1$ . Hence  $s$  does not meet both faces of the cube: either all  $s_i > 0$ , and then  $r \leq q$ , or some  $s_j = 0$ , all  $s_i < 1$ , and then  $r \geq q$ . Since  $\sum r_i = d = \sum q_i$ , in both cases  $r = q$ .

For the distance let  $w = e^s$ ,  $f = q - p$ , and let  $c$  be the midpoint of the range of  $z^2$ . Lemma 2.5 with  $r = q$  gives  $Lz \leq z \circ f$ , so, using  $\sum_i f_i = 0$ ,

$$z^T Lz \leq \sum_i z_i^2 f_i = \sum_i (z_i^2 - c) f_i \leq \frac{1}{2} \text{osc}(z^2) \|f\|_1.$$

As  $(z_i - z_j)^2 \geq 4 \min w^2 (w_i - w_j)^2$ , [Lemmas 2.2](#) and [2.3](#) give

$$\|P(s) - P\|_F^2 \leq \frac{2w^T Lw}{\min w^2} \leq \frac{z^T Lz}{2 \min z^2} \leq \frac{\text{osc}(z^2)}{4 \min z^2} \|f\|_1 \leq \frac{4^2 - 1}{4} \|f\|_1.$$

Finally apply [Lemma 2.1](#) to the whitened frame  $DU(U^T D^2 U)^{-1/2}$ ,  $D = \text{Diag}(w)$ , whose projection is  $P(s)$ .  $\square$

*Remark 2.9.* The hypothesis forces  $q_i \geq p_i/2 > 0$ : for  $n \geq 2$ , the barrier for  $S = [n] \setminus \{i\}$  gives  $1 \leq (L\psi)_i \leq p_i(1 - p_i)H$ , so  $\beta \leq 1/(2H) \leq p_i/2$ . The proof uses no existence theorem for scalings, no flow, and no genericity or connectivity argument: the box supplies a minimiser, the median barrier keeps it away from one face, and the trace identity  $\sum_i r_i = d$  does the rest. With  $\beta H < 1$  and a box of side  $T > -\log(1 - \beta H)$  the same proof gives scale ratio  $(1 - \beta H)^{-1}$  and cost  $\frac{1}{4}((1 - \beta H)^{-4} - 1)\|q - p\|_1$ .

## 2.4 Barriers from a dense core

**Lemma 2.10** (Dense core with exceptional vertices). *Let  $L$  be the Laplacian of nonnegative symmetric weights  $w_{ij}$  on  $[n]$ , and let  $[n] = \mathcal{G} \sqcup \mathcal{B}$ . Suppose:*

- (a) *there is  $\kappa > 0$  with  $\sum_{j \in S} w_{ij} \geq \kappa$  for every  $i \in \mathcal{G}$  and every  $S \subseteq [n] \setminus \{i\}$  with  $|S| \geq n/2$ ;*
- (b) *there are  $y \in \mathbb{R}^{\mathcal{B}}$ ,  $y \geq 0$ , and  $R, F_{\mathcal{B}} \geq 0$  with  $\|y\|_{\infty} \leq R$ ,  $(L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$  for  $i \in \mathcal{B}$ , and  $\sum_{j \in \mathcal{B}} w_{ij}y_j \leq F_{\mathcal{B}}$  for  $i \in \mathcal{G}$ . (If  $\mathcal{B} = \emptyset$  take  $R = F_{\mathcal{B}} = 0$ .)*

*Then  $H(L) \leq (1 + F_{\mathcal{B}})/\kappa + R$ . Moreover, with  $b = |\mathcal{B}|$ :*

- (i) (a) *holds with  $\kappa = \vartheta\gamma$  if  $\vartheta \in (0, \frac{1}{2}]$ ,  $\gamma > 0$ , and every  $i \in \mathcal{G}$  has at least  $(\frac{1}{2} + \vartheta)n$  indices  $j \neq i$  with  $w_{ij} \geq \gamma/n$ ;*
- (ii) (b) *holds with  $R = \sqrt{b}/\lambda$ ,  $F_{\mathcal{B}} = b$ , if  $x^T Lx \geq \lambda\|x\|^2$  for every  $x$  supported on  $\mathcal{B}$  ( $\lambda > 0$ );*
- (iii) *for  $w_{ij} = P_{ij}^2$ , (b) holds with  $R = \frac{8}{3\alpha}$ ,  $F_{\mathcal{B}} = \frac{2}{3\alpha} \max_i p_i$  if  $\alpha > 0$ ,  $p_i \geq \alpha/2$  on  $\mathcal{B}$  and  $\sum_{i \in \mathcal{B}} p_i \leq \frac{1}{4}$ .*

*Proof.* Let  $|S| \geq n/2$ , put  $\lambda_0 = (1 + F_{\mathcal{B}})/\kappa$ , extend  $y$  by 0 to  $\mathcal{G}$ , and let  $\psi = (\lambda_0 \mathbf{1} + y) \circ \mathbf{1}_{S^c}$ , so  $0 \leq \psi \leq \lambda_0 + R$  and  $\psi_j \leq \lambda_0 + y_j$  for all  $j$ . For  $i \in \mathcal{G} \setminus S$ ,  $\psi_i = \lambda_0$  and  $\psi_j = 0$  on  $S$ , so

$$(L\psi)_i \geq \lambda_0 \sum_{j \in S} w_{ij} - \sum_{j \in \mathcal{B}} w_{ij}y_j \geq \lambda_0 \kappa - F_{\mathcal{B}} = 1.$$

For  $i \in \mathcal{B} \setminus S$ ,  $\psi_i = \lambda_0 + y_i$ , so  $(L\psi)_i \geq \sum_j w_{ij}(y_i - y_j) = (L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$ . Thus  $\psi$  is an  $S$ -barrier.

(i) If  $N_i$  is the set of such  $j$  and  $S \subseteq [n] \setminus \{i\}$ , then  $|N_i \cap S| \geq |N_i| + |S| - (n - 1) > \vartheta n$ . (ii)  $L = L_{\mathcal{B}\mathcal{B}} \succeq \lambda I$ ; let  $y = L^{-1} \mathbf{1}_{\mathcal{B}}$ . If  $y$  had a negative minimum at  $i$ , then  $(Ly)_i = y_i \sum_{j \notin \mathcal{B}} w_{ij} + \sum_{j \in \mathcal{B}} w_{ij}(y_i - y_j) \leq 0$ , a contradiction; so  $y \geq 0$ , and  $\|y\|_{\infty} \leq \|y\|_2 \leq \sqrt{b}/\lambda$ . Since  $(L \mathbf{1}_{\mathcal{B}})_j = \sum_{i \in \mathcal{G}} w_{ij}$  for  $j \in \mathcal{B}$ , for each  $i_0 \in \mathcal{G}$

$$\sum_{j \in \mathcal{B}} w_{i_0 j} y_j \leq \sum_{i \in \mathcal{G}} \sum_{j \in \mathcal{B}} w_{ij} y_j = \mathbf{1}_{\mathcal{B}}^T L y = b.$$

(iii)  $P_{ij}^2 \leq p_i p_j$  gives, for  $i \in \mathcal{B}$ ,  $(L_{\mathcal{B}\mathcal{B}} \mathbf{1}_{\mathcal{B}})_i = p_i - \sum_{j \in \mathcal{B}} P_{ij}^2 \geq p_i(1 - \sum_{j \in \mathcal{B}} p_j) \geq \frac{3\alpha}{8}$ , and for  $i \in \mathcal{G}$ ,  $\sum_{j \in \mathcal{B}} P_{ij}^2 \leq p_i/4$ . Take  $y = \frac{8}{3\alpha} \mathbf{1}_{\mathcal{B}}$ .  $\square$

*Remark 2.11* (Poisson constants). The  $\ell_\infty$  Poisson constant  $K$  of  $L$  is the least  $K$  such that every  $f \perp \mathbf{1}$  has a solution of  $Lg = f$  with  $\|g\|_\infty \leq K\|f\|_\infty$ . Then  $K \leq H \leq 2K$ . Indeed, for  $|S| \geq n/2$  the forcing  $f = \mathbf{1} - \frac{n}{|S|}\mathbf{1}_S$  has  $\|f\|_\infty \leq 1$ , and  $g + K\mathbf{1}$  is an  $S$ -barrier; conversely, if  $H < \infty$  the graph is connected, so  $Lg = f$  is solvable for every  $f \perp \mathbf{1}$ , and if  $\|f\|_\infty \leq 1$  and  $m$  is a median of  $g$ , the argument of [Lemma 2.7](#) applied to  $g - m\mathbf{1} - \beta'\psi$  ( $\beta' > 1$ ) gives  $g \leq m + H$ , and symmetrically  $g \geq m - H$ . Barriers are more convenient than Poisson solutions because they can be written down explicitly, as in [Lemma 2.10](#).

## 2.5 From a seed to an exact frame

Both constructions produce the following object.

**Definition 2.12** (Seed). Let  $\Theta \geq 2$  and  $t > 0$ . A  $(\Theta, t)$ -seed for  $X \in \mathbb{R}^{n \times d}$  is a Parseval  $V \in \mathbb{R}^{n \times d}$  with

$$\|V - X\|_F^2 \leq \Theta t^2 d, \quad H(L_{VV^T}) \leq \frac{\Theta}{at^2}, \quad \max_i |(VV^T)_{ii} - a| \leq \frac{at^2}{2\Theta}.$$

**Corollary 2.13.** *If  $X$  has a  $(\Theta, t)$ -seed, there is an ENP frame  $W$  with  $\|X - W\|_F^2 \leq 3\Theta t^2 d$ .*

*Proof.* Apply [Theorem 2.8](#) to  $V$  with  $q = a\mathbf{1}$ :  $\beta H \leq \frac{1}{2}$ , and  $\|V - W\|_F^2 \leq \frac{15}{4}n \cdot \frac{at^2}{2\Theta} \leq t^2 d$ . Then  $\|X - W\|_F^2 \leq 2\|X - V\|_F^2 + 2\|V - W\|_F^2 \leq (2\Theta + 2)t^2 d \leq 3\Theta t^2 d$ .  $\square$

## 3 Bounded rank: the Hamilton–Moitra estimate

This section reproduces the radial-isotropic and ordered-coordinate argument of Hamilton and Moitra [\[2\]](#), with a direct compactness proof of the existence of the radial scaling. It is used for  $d$  below a large absolute constant, and it also shows that ENP frames exist.

**Lemma 3.1.** *Let  $1 \leq d \leq n$ , let  $X \in \mathbb{R}^{n \times d}$  have  $\|x_i\|^2 = a$  for all  $i$ , and let  $\eta = \|X^T X - I\|_{\text{op}}$ . There is an ENP frame  $W$  with  $\|X - W\|_F^2 \leq \eta d(d-1)$ .*

*Proof.* If  $d = 1$  then  $X^T X = na = 1$  and  $W = X$  works. Let  $d \geq 2$ .

*Step 1: full spark.* Suppose first that every  $d \times d$  minor  $\mu_S = \det(X_S)^2$  ( $S \subset [n]$ ,  $|S| = d$ ) is nonzero. Consider  $F(s) = \frac{1}{2} \log \det(X^T \text{Diag}(e^{2s})X) - a \sum_i s_i$ . By Cauchy–Binet,

$$F(s) = \frac{1}{2} \log \sum_{|S|=d} \mu_S e^{T(s)_S}, \quad T(s)_S = 2 \left( \sum_{i \in S} s_i - a \sum_i s_i \right),$$

where  $T : \mathbb{R}^n \rightarrow \mathbb{R}^{\binom{n}{d}}$  is linear and  $\sum_S T(s)_S = 0$  because each  $i$  lies in a fraction  $d/n = a$  of the  $d$ -sets. The function  $\mathcal{E}(y) = \sum_S \mu_S e^{y_S}$  restricted to the subspace  $\text{im } T$  has the nonempty sublevel set  $\{\mathcal{E} \leq \mathcal{E}(0)\}$ , on which  $y_S \leq \log(\mathcal{E}(0)/\mu_S)$  for every  $S$  and hence, by  $\sum_S y_S = 0$ , also  $y_S$  is bounded below. This set is compact, so  $\mathcal{E}|_{\text{im } T}$  attains its minimum at some  $T(s^*)$ , and  $s^*$  minimises  $F$ .

Let  $D = \text{Diag}(e^{s^*})$ ,  $M = (X^T D^2 X)^{-1/2}$  and  $W = DXM$ . The computation in the proof of [Lemma 2.4](#) (which does not use  $X^T X = I$ ) gives  $\partial_i F = \|w_i\|^2 - a$ , so  $W$  is ENP, and since  $w_i = e^{s_i^*} M x_i$  has norm  $\sqrt{a}$ ,

$$w_i = \sqrt{a} \frac{M x_i}{\|M x_i\|}.$$

*Step 2: ordered coordinates.* Replacing  $X, W$  by  $XO, WO$  for a suitable  $O \in O(d)$  changes neither the hypotheses nor the distance, and makes  $M = \text{Diag}(\lambda_1, \dots, \lambda_d)$  with  $0 < \lambda_1 \leq \dots \leq \lambda_d$ .

Then  $w_{ij} = \lambda_j x_{ij} / \sqrt{c_i}$  with  $c_i = a^{-1} \sum_\ell \lambda_\ell^2 x_{i\ell}^2$ , so  $w_{ij}$  has the sign of  $x_{ij}$  and  $w_{ij}^2 - x_{ij}^2 = x_{ij}^2 (\lambda_j^2 / c_i - 1)$  is nonpositive and then nonnegative as  $j$  increases. As  $\sum_j (x_{ij}^2 - w_{ij}^2) = a - a = 0$ , the prefix sums  $\Delta_i(k) = \sum_{j \leq k} (x_{ij}^2 - w_{ij}^2)$  are nonnegative, vanish at  $k = d$ , and

$$\sum_j |x_{ij}^2 - w_{ij}^2| = 2 \max_k \Delta_i(k) \leq 2 \sum_{k=1}^{d-1} \Delta_i(k).$$

Since  $W^T W = I$  and  $(X^T X)_{jj} \leq 1 + \eta$ ,  $\sum_i \Delta_i(k) = \sum_{j \leq k} ((X^T X)_{jj} - 1) \leq k\eta$ . Equal signs give  $(x_{ij} - w_{ij})^2 \leq |x_{ij}^2 - w_{ij}^2|$ , and therefore

$$\|X - W\|_F^2 \leq 2\eta \sum_{k=1}^{d-1} k = \eta d(d-1).$$

*Step 3: general  $X$ .* Let  $G$  be a Vandermonde matrix with rows  $(1, \theta_i, \dots, \theta_i^{d-1})$  for distinct  $\theta_i$ ; all its maximal minors are nonzero. Each maximal minor of  $X + hG$  is a polynomial in  $h$  with nonzero leading coefficient, so for all but finitely many  $h$  the matrix  $X + hG$  is full spark; choose such  $h_k \rightarrow 0$  (so small that no row of  $X + h_k G$  vanishes) and normalise each row of  $X + h_k G$  to length  $\sqrt{a}$  (a positive diagonal left factor, which preserves full spark). The resulting  $X_k \rightarrow X$  have ENP frames  $W_k$  with  $\|X_k - W_k\|_F^2 \leq \eta_k d(d-1)$ ,  $\eta_k \rightarrow \eta$ . The set of ENP frames is compact; a limit point  $W$  of  $(W_k)$  is ENP and satisfies the bound.  $\square$

**Corollary 3.2.** *For all  $1 \leq d \leq n$  an ENP frame exists. Consequently every equal-norm  $X \in \mathbb{R}^{n \times d}$  is within  $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2\|W\|_F^2 = 4d$  of an ENP frame  $W$ .*

*Proof.* Apply Lemma 3.1 to  $x_i = \sqrt{a} e_1$  for all  $i$ .  $\square$

## 4 The many-row seed

**Theorem 4.1.** *There are absolute constants  $B, c_*, C_* > 0$  such that the following holds. Let  $2 \leq d$ ,  $n \geq Bd^2$ , let  $X \in \mathbb{R}^{n \times d}$  have  $\|x_i\|^2 = a = d/n$  for all  $i$ , and suppose  $\eta := \|X^T X - I\|_{\text{op}}$  satisfies  $0 < \eta \leq c_*$ . Then there is an ENP frame  $W$  with  $\|X - W\|_F^2 \leq C_* \eta d$ .*

Throughout,  $S = X^T X$ , so  $\|S - I\|_{\text{op}} \leq \eta \leq c_* \leq \frac{1}{2}$  (the choice of  $c_*$  below gives  $c_* \leq \frac{1}{64}$ ), and  $e_i = x_i / \sqrt{a}$ .

### 4.1 The noise

In the Frobenius space  $\mathbb{R}^{n \times d}$  consider the subspaces

$$\mathcal{E} = \{Z : \langle x_i, z_i \rangle = 0 \ \forall i\}, \quad \mathcal{F} = \{Z \in \mathcal{E} : X^T Z + Z^T X = 0\},$$

and let  $m = \dim \mathcal{E} - \dim \mathcal{F} \leq \dim \text{Sym}_d \leq d^2 \leq n$ . For a standard Gaussian  $g \in \mathbb{R}^{n \times d}$  put

$$Z_0 = n^{-1/2} \Pi_{\mathcal{E}} g, \quad Z = n^{-1/2} \Pi_{\mathcal{F}} g, \quad Z_\perp = Z_0 - Z = n^{-1/2} \Pi_{\mathcal{E} \ominus \mathcal{F}} g.$$

$Z$  and  $Z_\perp$  are independent. The rows  $z_{0,i} = n^{-1/2} (I - e_i e_i^T) g_i$  of  $Z_0$  are independent. Since  $\text{Cov}(Z) \preceq \text{Cov}(Z_0)$ , each row  $z_i$  of  $Z$  is Gaussian with covariance  $\Sigma_i \preceq n^{-1} (I - e_i e_i^T)$ , and  $\text{Cov}(\text{vec } Z) \preceq I/n$ . Moreover

$$\mathbb{E} \|Z\|_F^2 = \frac{n(d-1) - m}{n} \leq d, \quad \mathbb{E} \|Z - Z_0\|_F^2 = \frac{m}{n} \leq \frac{d^2}{n}. \quad (4.1)$$

**Lemma 4.2** (Row moments). *Let  $r_i = \|z_i\|^2/a$ ,  $\mu_i = \mathbb{E}r_i$  and  $\delta_i = 1 - \mu_i$ . Then  $1/d \leq \delta_i \leq 1$ ,  $\sum_i a\delta_i = 1 + m/n \leq 2$ , and*

$$\mathcal{M} := a \sum_i \mathbb{E}[(r_i - 1)^2(1 + r_i)^2] \leq C_{\mathcal{M}},$$

with  $C_{\mathcal{M}}$  absolute.

*Proof.*  $\tilde{\Sigma}_i = \Sigma_i/a \preceq d^{-1}I$  has trace  $\mu_i \leq (d-1)/d$ , so  $\text{tr } \tilde{\Sigma}_i^2 \leq 1/d$  and  $\text{tr } \tilde{\Sigma}_i^4 \leq d^{-3}$ . For  $\xi_i = r_i - \mu_i$  the Gaussian moment formulas give  $\mathbb{E}\xi_i^2 = 2 \text{tr } \tilde{\Sigma}_i^2 \leq 2/d$  and  $\mathbb{E}\xi_i^4 = 12(\text{tr } \tilde{\Sigma}_i^2)^2 + 48 \text{tr } \tilde{\Sigma}_i^4 \leq 60/d^2$ . Next  $\sum_i a\delta_i = d - \mathbb{E}\|Z\|_F^2 = 1 + m/n$  by (4.1). Finally  $(r_i - 1)^2 \leq 2\xi_i^2 + 2\delta_i^2$  and  $(1 + r_i)^2 \leq 8 + 2\xi_i^2$ , so

$$\mathbb{E}(r_i - 1)^2(1 + r_i)^2 \leq 16\mathbb{E}\xi_i^2 + 4\mathbb{E}\xi_i^4 + 16\delta_i^2 + 4\delta_i^2\mathbb{E}\xi_i^2 \leq \frac{32}{d} + \frac{240}{d^2} + \left(16 + \frac{8}{d}\right)\delta_i^2,$$

and, as  $\delta_i^2 \leq \delta_i$  and  $na = d$ ,  $\mathcal{M} \leq 32 + \frac{240}{d} + (16 + \frac{8}{d})a \sum_i \delta_i \leq 32 + 120 + 2 \cdot 20 = 192$  for  $d \geq 2$ ; so  $C_{\mathcal{M}} = 192$  works.  $\square$

## 4.2 Row normalisation and the exact second-order identity

For  $0 < t \leq 1$  set

$$v_i = \frac{x_i + tz_i}{\sqrt{1 + t^2 r_i}}, \quad v_{0,i} = \frac{x_i + tz_{0,i}}{\sqrt{1 + t^2 r_{0,i}}}, \quad r_{0,i} = \|z_{0,i}\|^2/a.$$

Since  $z_i \perp x_i$ ,  $\|v_i\|^2 = \|v_{0,i}\|^2 = a$  exactly. The map  $y \mapsto \sqrt{a}y/\|y\|$  is the metric projection onto the closed ball of radius  $\sqrt{a}$  on  $\{\|y\| \geq \sqrt{a}\}$ , hence 1-Lipschitz there. Thus

$$\|V - X\|_F \leq t\|Z\|_F, \quad \|V - V_0\|_F \leq t\|Z - Z_0\|_F, \quad \|V\|_{\text{op}} \leq \|X + tZ\|_{\text{op}}, \quad \|V_0\|_{\text{op}} \leq \|X + tZ_0\|_{\text{op}}, \quad (4.2)$$

the last two because  $V = \text{Diag}(\lambda)(X + tZ)$  with  $0 < \lambda_i \leq 1$ , and similarly for  $V_0$ .

Expanding  $1/(1 + t^2 r_i) = 1 - t^2 c_i$  with

$$c_i = \frac{r_i}{1 + t^2 r_i},$$

and using  $X^T Z + Z^T X = 0$ , one finds the exact identity

$$V^T V = S + t^2 Q(Z) + R_3 + R_4, \quad Q(Z) = Z^T Z - \sum_i r_i x_i x_i^T, \quad (4.3)$$

$$R_3 = -t^3 \sum_i c_i (x_i z_i^T + z_i x_i^T), \quad R_4 = -t^4 \sum_i c_i (z_i z_i^T - r_i x_i x_i^T).$$

**Lemma 4.3** (Mean and fluctuation of  $Q$ ).  $\|\mathbb{E}Q(Z) - (I - S)\|_{\text{op}} \leq m/n$  and  $\mathbb{E}\|Q(Z) - \mathbb{E}Q(Z)\|_F^2 \leq 8d^2/n$ .

*Proof.* For  $Z_0$ :  $\mathbb{E}Z_0^T Z_0 = n^{-1} \sum_i (I - e_i e_i^T) = I - S/d$  and  $\mathbb{E}r_{0,i} = (d-1)/d$ , so  $\mathbb{E}Q(Z_0) = I - S/d - (1 - 1/d)S = I - S$ . As  $Z_0 = Z + Z_{\perp}$  with independent centred summands and  $Q$  is quadratic,  $\mathbb{E}Q(Z) = \mathbb{E}Q(Z_0) - \mathbb{E}Q(Z_{\perp})$ , and  $\mathbb{E}Q(Z_{\perp})$  is the difference of the positive semidefinite matrices  $\mathbb{E}Z_{\perp}^T Z_{\perp}$  and  $\sum_i \mathbb{E}(\|z_{\perp,i}\|^2/a)x_i x_i^T$ , both of trace  $\mathbb{E}\|Z_{\perp}\|_F^2 = m/n$ ; so its norm is at most  $m/n$ .

Entry  $(\alpha, \beta)$  of  $Q$  is the quadratic form  $\sum_i z_i^T \mathbf{B}_{\alpha\beta,i} z_i$  with  $\mathbf{B}_{\alpha\beta,i} = \frac{1}{2}(E_{\alpha\beta} + E_{\beta\alpha}) - (e_i)_\alpha (e_i)_\beta I$ , where the  $E_{\alpha\beta}$  are the matrix units of  $\mathbb{R}^{d \times d}$ . For a Gaussian vector with covariance  $\Sigma \preceq I/n$  and symmetric coefficient matrix  $\mathbf{B}$ , the variance of the quadratic form is  $2\|\Sigma^{1/2} \mathbf{B} \Sigma^{1/2}\|_F^2 \leq 2\|\mathbf{B}\|_F^2/n^2$ ; we apply this to  $\text{vec } Z$  and the block-diagonal matrix with blocks  $\mathbf{B}_{\alpha\beta,i}$ . Since  $\sum_{\alpha,\beta} \|\mathbf{B}_{\alpha\beta,i}\|_F^2 \leq 2d^2 + 2d \leq 4d^2$  for each  $i$ , summing over  $\alpha, \beta$  and the  $n$  blocks gives the claim.  $\square$

**Lemma 4.4** (Remainder). *Let  $\bar{c} = 1/(1+t^2)$ ,  $\Gamma = \text{Diag}(c_i - \bar{c})$  and  $\Lambda = \text{Diag}(c_i r_i - \bar{c})$ . Then*

$$\|R_3\|_{\text{op}} \leq 2t^3 \|X\|_{\text{op}} \|\Gamma Z\|_F, \quad \|R_4\|_{\text{op}} \leq t^4 (\|Z\|_{\text{op}}^2 + \|Z\|_{\text{op}} \|\Gamma Z\|_F + \|S\|_{\text{op}} + \|X\|_{\text{op}} \|\Lambda X\|_F),$$

and  $\mathbb{E}\|\Gamma Z\|_F^2 \leq \mathcal{M}$ ,  $\mathbb{E}\|\Lambda X\|_F^2 \leq \mathcal{M}$ .

*Proof.* Because  $\sum_i (x_i z_i^T + z_i x_i^T) = X^T Z + Z^T X = 0$ , the constant part  $\bar{c}$  of the weights drops out of  $R_3$ :  $R_3 = -t^3(X^T \Gamma Z + Z^T \Gamma X)$ , which gives the first bound. Next  $\sum_i c_i z_i z_i^T = \bar{c} Z^T Z + Z^T \Gamma Z$  and  $\sum_i c_i r_i x_i x_i^T = \bar{c} S + X^T \Lambda X$ , with  $\|Z^T \Gamma Z\|_{\text{op}} \leq \|Z\|_{\text{op}} \|\Gamma Z\|_F$  and similarly for  $\Lambda$ , and  $\bar{c} \leq 1$ . For the moments,

$$c_i - \bar{c} = \frac{r_i - 1}{(1+t^2 r_i)(1+t^2)}, \quad c_i r_i - \bar{c} = \frac{(r_i - 1)(1+r_i+t^2 r_i)}{(1+t^2 r_i)(1+t^2)},$$

so  $|c_i - \bar{c}| \leq |r_i - 1|$  and, as  $1+r+t^2 r \leq (1+r)(1+t^2 r)$ ,  $|c_i r_i - \bar{c}| \leq |r_i - 1|(1+r_i)$ . Hence  $\|\Gamma Z\|_F^2 = a \sum_i (c_i - \bar{c})^2 r_i \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$  and  $\|\Lambda X\|_F^2 = a \sum_i (c_i r_i - \bar{c})^2 \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$ ; take expectations and use Lemma 4.2.  $\square$

*Remark 4.5.* The termwise bound  $\|R_3\|_{\text{op}} \leq 2t^3 \sum_i c_i \|x_i\| \|z_i\| \approx 2t^3 d$  would only allow  $\eta \lesssim d^{-2}$ . Centring the weights at  $\bar{c}$ , which is allowed by the first-order constraint, replaces the  $n$  terms of size  $a$  by fluctuations of relative size  $d^{-1/2}$ , whose  $\ell_2$ -sum over the rows is  $O(1)$ .

### 4.3 A dense reference graph

**Lemma 4.6.** *There are absolute  $h, t_0 \in (0, 1]$  and  $c_1 > 0$  such that for  $d \geq 2$  and  $t \leq t_0$ , with probability at least  $1 - ne^{-c_1(n-1)}$ , every  $i$  has at least  $0.9(n-1)$  indices  $j \neq i$  with*

$$|\langle v_{0,i}, v_{0,j} \rangle| \geq h t \sqrt{a/n}.$$

*Proof.* Write  $z_{0,j}/\sqrt{a} = d^{-1/2} g'_j$  with  $g'_j \sim N(0, I - e_j e_j^T)$  independent. Fix  $i$  and condition on row  $i$ ; let  $f = v_{0,i}/\sqrt{a}$ , a unit vector. For  $j \neq i$ ,

$$\frac{\langle v_{0,i}, v_{0,j} \rangle}{a} = \frac{\nu_j}{N_j}, \quad \nu_j = u_j + \frac{t}{\sqrt{d}} \langle f, g'_j \rangle, \quad u_j = \langle f, e_j \rangle, \quad N_j = (1 + \frac{t^2}{d} \|g'_j\|^2)^{1/2} \geq 1,$$

and the target inequality reads  $|\nu_j|/N_j \geq ht/\sqrt{d}$ . By Markov,  $\mathbb{P}(N_j > 2) \leq t^2/3$ . On  $\{N_j \leq 2\}$  failure requires  $|\nu_j| < 2ht/\sqrt{d}$ . Here  $\nu_j \sim N(u_j, t^2(1 - u_j^2)/d)$ . If  $|u_j| \leq \frac{1}{2}$  the variance is at least  $3t^2/(4d)$  and the Gaussian density bound gives probability at most  $4h/\sqrt{3\pi/2} \leq 3h$ . If  $|u_j| > \frac{1}{2}$  and  $2ht \leq \frac{1}{4}$ , failure requires  $\frac{t}{\sqrt{d}} |\langle f, g'_j \rangle| > \frac{1}{4}$ , of probability at most  $16t^2/d$ . Choose  $h$  and  $t_0$  so that  $3h + t_0^2/3 + 8t_0^2 \leq \frac{1}{50}$ , for instance  $h = \frac{1}{300}$  and  $t_0 = \frac{1}{30}$  (this also gives  $2ht \leq \frac{1}{4}$ , and  $16t^2/d \leq 8t^2$  as  $d \geq 2$ ). The failure indicators for  $j \neq i$  are conditionally independent with probabilities at most  $\frac{1}{50}$ , so by the exponential Markov inequality (with parameter 1) more than  $0.1(n-1)$  failures have probability at most  $\exp(-(n-1)[0.1 - \frac{e-1}{50}]) \leq e^{-c_1(n-1)}$ . Take a union bound over  $i$ .  $\square$

#### 4.4 Proof of Theorem 4.1

*Constants.* Fix  $h, t_0, c_1$  from Lemma 4.6. Put  $\kappa_0 = h^2/200$ ,  $C' = 36^2 \cdot 20$ , and  $C_{\text{mix}} = 125/h^2 + 3$ , then  $\Theta = \max\{42, C_{\text{mix}}\}$  and  $\theta = \min\{1/(4\Theta), h/60\}$ . Let  $C_R = 300 + 100\sqrt{C_{\mathcal{M}}}$ , choose  $t_1 = \min\{t_0, \theta/(16C_R)\}$ , and set

$$t^2 = 4\eta/\theta, \quad c_* = \theta t_1^2/4,$$

so that  $\eta \leq c_*$  gives  $t \leq t_1$ . Finally choose  $B$  so large that  $n \geq Bd^2$  implies  $d^2/n \leq \theta/4$ ,  $\sqrt{160d^2/n} \leq \theta/4$ ,  $2aC'd/\kappa_0 \leq \frac{1}{4}$  (this bounds  $\sum_{i \in \mathcal{B}_0} 2a$  for the set  $\mathcal{B}_0$  below, which has  $|\mathcal{B}_0| \leq C'd/\kappa_0$ ),  $n \geq 30$ , and  $ne^{-c_1(n-1)} \leq 0.01$ .

*The event.* By Markov's inequality, (4.1), Lemmas 4.3 and 4.4, the net bound (Lemma A.1(c)) for  $Z$  and  $Z_0$ , and Lemma 4.6, with positive probability all of the following hold:

$$\|Z\|_{\text{F}}^2 \leq 20d, \quad \|Z - Z_0\|_{\text{F}}^2 \leq \frac{20d^2}{n}, \quad \|Q - \mathbb{E}Q\|_{\text{F}}^2 \leq \frac{160d^2}{n}, \quad \|\Gamma Z\|_{\text{F}}^2 \leq 20\mathcal{M}, \quad \|\Lambda X\|_{\text{F}}^2 \leq 20\mathcal{M},$$

$\|Z\|_{\text{op}}, \|Z_0\|_{\text{op}} \leq 16$ , and the conclusion of Lemma 4.6: the five Markov events fail with probability at most  $\frac{1}{20}$  each, the two net bounds (Lemma A.1(c) with  $\sigma^2 = 1/n$ ,  $u = 16$ ) with probability at most  $2 \cdot 5^{n+d}e^{-8n}$  each, and Lemma 4.6 with probability at most 0.01. Fix such a sample.

*Spectral error.* By (4.3),  $V^T V - I = (1 - t^2)(S - I) + t^2(\mathbb{E}Q - (I - S)) + t^2(Q - \mathbb{E}Q) + R_3 + R_4$ . Since  $\|X\|_{\text{op}} \leq \sqrt{2}$ ,  $\|S\|_{\text{op}} \leq \frac{3}{2}$  and  $t \leq 1$ , Lemma 4.4 and the event give  $\|R_3\|_{\text{op}} + \|R_4\|_{\text{op}} \leq t^3(2\sqrt{40\mathcal{M}} + 256 + 16\sqrt{20\mathcal{M}} + \frac{3}{2} + \sqrt{40\mathcal{M}}) \leq C_R t^3$ , and therefore

$$\delta := \|V^T V - I\|_{\text{op}} \leq \eta + t^2 \frac{d^2}{n} + t^2 \sqrt{\frac{160d^2}{n}} + C_R t^3 \leq \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{16} t^2 < \theta t^2.$$

*Graph.* By (4.2),  $\|V\|_{\text{op}}, \|V_0\|_{\text{op}} \leq \sqrt{2} + 16 \leq 18$  and  $\|VV^T - V_0V_0^T\|_{\text{F}} \leq 36t\|Z - Z_0\|_{\text{F}}$ , so  $\|VV^T - V_0V_0^T\|_{\text{F}}^2 \leq C't^2d^2/n$ . Let  $\mathcal{B}_0$  be the set of rows whose contribution to this sum exceeds  $\kappa_0 at^2$ ; then  $|\mathcal{B}_0| \leq C'd/\kappa_0$ . A row  $i \notin \mathcal{B}_0$  has at most  $\kappa_0 at^2/((h/2)^2 t^2 a/n) = n/50$  indices with  $|(VV^T - V_0V_0^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$ , so by Lemma 4.6 it has at least  $0.85n$  indices  $j \neq i$  with  $|(VV^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$ .

Let  $U = V(V^T V)^{-1/2}$ ,  $P = UU^T$ . By Lemma 2.1 (rows of  $P - VV^T$ ) a row loses at most  $6a\theta^2 t^4/((h/4)^2 t^2 a/n) \leq n/20$  of these indices, so each  $i \notin \mathcal{B}_0$  has at least  $0.8n$  indices  $j \neq i$  with  $P_{ij}^2 \geq (h^2/16)at^2/n$ . Also  $a/2 \leq P_{ii} \leq 2a$  and  $\sum_{i \in \mathcal{B}_0} P_{ii} \leq \frac{1}{4}$ . Lemma 2.10 with  $\mathcal{B} = \mathcal{B}_0$ , part (i) with  $\vartheta = 0.3$  and  $\gamma = h^2 at^2/16$ , and part (iii) with  $\alpha = a$  (so  $R = \frac{8}{3a}$ ,  $F_{\mathcal{B}} \leq \frac{4}{3}$ ) shows that  $H(L_P) \leq \frac{7/3}{0.3\gamma} + \frac{8}{3a} \leq C_{\text{mix}}/(at^2)$ .

*Seed.*  $\|U - X\|_{\text{F}}^2 \leq 2d\delta^2 + 2t^2\|Z\|_{\text{F}}^2 \leq 42t^2d$ , the diagonal error is at most  $2a\delta \leq 2\theta at^2 \leq at^2/(2\Theta)$ , and the barrier constant is at most  $\Theta/(at^2)$ . So  $U$  is a  $(\Theta, t)$ -seed, and Corollary 2.13 gives an ENP frame within  $3\Theta t^2 d = (12\Theta/\theta)\eta d$  of  $X$ ; thus  $C_* = 12\Theta/\theta$  works.  $\square$

## 5 The moderate-row seed

**Theorem 5.1.** *There are absolute constants  $A, C_m, \varepsilon_0 > 0$  such that the following holds. Let  $U \in \mathbb{R}^{n \times d}$  be Parseval with  $2d \leq n$ ,  $d \geq A \log(2n)$ , and  $|||u_i||^2 - a| \leq \varepsilon a$  for all  $i$ , where  $0 < \varepsilon \leq \varepsilon_0$ . Then there is an ENP frame  $\hat{U}$  with  $\|\hat{U} - U\|_{\text{F}}^2 \leq C_m \varepsilon d$ .*

Throughout this section  $P = UU^T$ ,  $p = \text{diag } P$ ,  $D_p = \text{Diag}(p)$ ,  $L = L_P$ , and  $\varepsilon_0 \leq \frac{1}{2}$ , so  $\frac{a}{2} \leq p_i \leq \frac{3a}{2} \leq \frac{3}{4}$ . For a symmetric  $M \in \mathbb{R}^{n \times n}$ ,  $\mathcal{L}(M) = \sum_{i < j} M_{ij}^2 (e_i - e_j)(e_i - e_j)^T$  is the

Laplacian with weights  $M_{ij}^2$ ; thus  $L = \mathcal{L}(P)$ ,  $x^T \mathcal{L}(M)x = \frac{1}{2} \|[\text{Diag } x, M]\|_F^2$ , and

$$\mathcal{L}(M) \preceq 2 \max_i \|M_i\|^2 I, \quad \mathcal{L}(M + M') \succeq \frac{1}{2} \mathcal{L}(M) - \mathcal{L}(M'), \quad (5.1)$$

by  $(x_i - x_j)^2 \leq 2x_i^2 + 2x_j^2$  and  $(v + w)^2 \geq \frac{1}{2}v^2 - w^2$ . We write  $K = nI - \mathbf{1}\mathbf{1}^T$  for the Laplacian of the complete graph.

**Scales.** Put  $\sigma = at^2$ , where  $t \asymp \sqrt{\varepsilon}$  is the size of the perturbation. The perturbation creates edges of weight  $\gtrsim \sigma/n$ , the resulting barrier constant is  $O(1/\sigma)$ , and so all diagonal errors must be at most a small multiple of  $\sigma$ . The table lists the errors and how each is controlled ( $\rho = D^2 t^2$  is the filter level,  $D$  a large constant).

| diagonal error        | size/ $\sigma$        | reason                                       |
|-----------------------|-----------------------|----------------------------------------------|
| input                 | $\varepsilon/t^2$     | $t^2 = K_0 \varepsilon$ , $K_0$ large        |
| linear term $2tAZ$    | $6D\sqrt{\log(2n)/d}$ | filter (Lemma 5.3), $d \geq A \log 2n$       |
| quadratic fluctuation | $\eta'$               | concentration, $d \geq A \log 2n$            |
| filter bias $t^2 b_*$ | 0                     | cancelled by the drift (Lemmas 5.6 and 5.10) |
| drift cross terms     | $40/D + 1/(2D^2)$     | $D$ large                                    |
| base bias $t^2 b_0$   | $12\varepsilon/d$     | Lemma 5.6                                    |
| retraction remainder  | $C_E t$               | $t$ small                                    |

### 5.1 The horizontal tangent space and the filtered covariance

Fix an isometry  $V \in \mathbb{R}^{n \times (n-d)}$  onto  $(\text{im } U)^\perp$ , and let  $\mathcal{Z} = \mathbb{R}^{(n-d) \times d}$  with the Frobenius inner product. For  $Z \in \mathcal{Z}$  the matrix  $VZ$  is horizontal ( $U^T VZ = 0$ ), and we put

$$Y_Z = VZU^T + UZ^T V^T, \quad \|(Y_Z)_i\|^2 = \|Z^T v_i\|^2 + \|Zu_i\|^2 \leq \|Z\|_{\text{op}}^2, \quad \|Y_Z\|_F^2 = 2\|Z\|_F^2.$$

Define

$$\mathcal{A} : \mathcal{Z} \rightarrow \mathbb{R}^n, \quad (\mathcal{A}Z)_i = v_i^T Z u_i = (VZU^T)_{ii}, \quad \mathcal{A}^* x = N_x := V^T \text{Diag}(x)U.$$

Then  $\mathcal{A}\mathcal{A}^* = L$ , i.e.  $\|N_x\|_F^2 = x^T Lx$  (Lemma 2.2). Let

$$\bar{\mathcal{A}} = D_p^{-1/2} \mathcal{A}, \quad S = \bar{\mathcal{A}}\bar{\mathcal{A}}^* = D_p^{-1/2} L D_p^{-1/2}, \quad \Omega = \bar{\mathcal{A}}^* \bar{\mathcal{A}}.$$

Since  $L \succeq 0$  and  $S = I - D_p^{-1/2}(P \circ P)D_p^{-1/2}$  with  $P \circ P \succeq 0$  (Schur product theorem),

$$0 \preceq S \preceq I, \quad 0 \preceq \Omega \preceq I, \quad \text{tr}(I - S) = \sum_i \frac{P_{ii}^2}{p_i} = d. \quad (5.2)$$

Fix an orthonormal eigenbasis of  $S$ , and for each of its members  $y$  with eigenvalue  $\mu > 0$  put  $f_y = D_p^{-1/2} y$  and  $Z_y = \mu^{-1/2} \bar{\mathcal{A}}^* y = \mu^{-1/2} N_{f_y}$ . Since  $\Omega Z_y = \mu Z_y$  and  $\langle Z_y, Z_{y'} \rangle = \langle y, Sy' \rangle / \sqrt{\mu \mu'}$ , the  $Z_y$  form an orthonormal eigenbasis of  $\Omega$  on  $(\ker \Omega)^\perp$ . Sums  $\sum_{\mu > 0}$  below run over these basis members (with multiplicity). Since  $p_i \geq a/2$ ,

$$\|f_y\|_\infty \leq \sqrt{2/a}. \quad (5.3)$$

**Definition 5.2** (Filtered covariance). For  $\rho > 0$  let  $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$  on  $\mathcal{Z}$ , let  $Z \sim N(0, C_\rho/n)$ , and  $Y = Y_Z$ .

**Lemma 5.3.** (a)  $0 \preceq C_\rho \preceq I$ , and  $C_\rho = I$  on  $\ker \Omega$ .

(b)  $\text{Cov}(\mathcal{A}Z) = n^{-1}D_p^{1/2} \rho S(I - S)(\rho I + S)^{-1}D_p^{1/2} \preceq (\rho/n)D_p$ .

(c)  $I - C_\rho = \Omega + R_\rho$  with  $R_\rho = \sum_{\mu>0} r_\mu Z_y \otimes Z_y$ ,  $r_\mu = \mu(1 - \mu)/(\rho + \mu) \geq 0$ , and

$$\sum_{\mu>0} r_\mu \leq d, \quad \sum_{\mu>0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{d}{2\sqrt{\rho}}, \quad \sum_{\mu>0} \frac{r_\mu}{\mu} \leq \frac{d}{\rho}.$$

*Proof.* (a) The eigenvalues of  $C_\rho$  are  $\rho(1 - \mu)/(\rho + \mu) \in [0, 1]$ . (b) By singular value calculus  $\mathcal{A}C_\rho\mathcal{A}^* = \rho S(I - S)(\rho I + S)^{-1}$ , whose eigenvalues  $\rho\mu(1 - \mu)/(\rho + \mu)$  are at most  $\rho$ . (c) On the  $\mu$ -eigenspace,  $1 - \rho(1 - \mu)/(\rho + \mu) = \mu + \mu(1 - \mu)/(\rho + \mu)$ . The sums follow from  $\mu/(\rho + \mu) \leq 1$ ,  $\sqrt{\mu}/(\rho + \mu) \leq 1/(2\sqrt{\rho})$ ,  $1/(\rho + \mu) \leq 1/\rho$ , and  $\sum_{\mu>0}(1 - \mu) \leq \text{tr}(I - S) = d$ .  $\square$

*Remark 5.4.* Only these properties of the filter  $\phi(\mu) = \rho(1 - \mu)/(\rho + \mu)$  are used:  $\phi(0) = 1$  (so  $C_\rho = I$  on  $\ker \Omega$ ),  $0 \leq \phi(\mu) \leq 1 - \mu$ ,  $\mu\phi(\mu) \leq \rho$ , and the sums in (c). The hard cutoff  $(1 - \mu)\mathbf{1}[\mu < \rho]$  satisfies them with  $\sum r_\mu/\sqrt{\mu} \leq d/\sqrt{\rho}$ , which doubles some constants below; the rational filter keeps every object a rational function of  $S$ , which is convenient for formalisation.

## 5.2 Commutator identities

For  $Z \in \mathcal{Z}$  define the quadratic diagonal and its polarisation

$$q_i(Z) = \|Z^T v_i\|^2 - \|Z u_i\|^2, \quad q_i(Z, M) = \langle Z^T v_i, M^T v_i \rangle - \langle Z u_i, M u_i \rangle,$$

so that  $q(Z + M) = q(Z) + 2q(Z, M) + q(M)$  and, by Cauchy–Schwarz,

$$|q_i(Z, M)| \leq \|(Y_Z)_i\| \|(Y_M)_i\|, \quad |q_i(M)| \leq \|(Y_M)_i\|^2. \quad (5.4)$$

**Lemma 5.5.** Let  $f \in \mathbb{R}^n$ ,  $F = \text{Diag}(f)$ , and  $R = I - 2P$  (an orthogonal matrix). Then:

(a)  $q(N_f) = \mathcal{A}\Gamma_f$  with  $\Gamma_f = V^T F R F U$ , and  $\|\Gamma_f\|_F \leq 2\|f\|_\infty \|N_f\|_F$ ;

(b)  $T_f := Y_{N_f} = F P + P F - 2 P F P$  satisfies  $[X, T_f] = R F [X, P] + [X, P] F R$  for diagonal  $X$ ; hence  $\mathcal{L}(T_f) \preceq 4\|f\|_\infty^2 L$ , and  $x^T \mathcal{L}(T_{e_k}) x \leq 2 \sum_j P_{kj}^2 (x_k - x_j)^2$ .

*Proof.* (a) For  $x \in \mathbb{R}^n$  put  $X = \text{Diag}(x)$ ,  $B_x = V^T X V$ ,  $A_x = U^T X U$ . Then  $\langle x, q(Z) \rangle = \langle Z, B_x Z - Z A_x \rangle$  for every  $Z$ , and, since  $X$  and  $F$  commute,

$$B_x N_f - N_f A_x = V^T [X(I - P)F - F P X] U = V^T [F(I - P)X - X P F] U = B_f N_x - N_x A_f.$$

Hence  $\langle x, q(N_f) \rangle = \langle N_f, B_f N_x - N_x A_f \rangle = \langle B_f N_f - N_f A_f, N_x \rangle$ . As  $\langle M, N_x \rangle = \sum_i x_i v_i^T M u_i = \langle \mathcal{A}M, x \rangle$  for every  $M$ , this says  $q(N_f) = \mathcal{A}(B_f N_f - N_f A_f)$ , and  $B_f N_f - N_f A_f = V^T F(I - P)F U - V^T F P F U = \Gamma_f$ . The norm bound follows from  $\|B_f\|_{\text{op}}, \|A_f\|_{\text{op}} \leq \|f\|_\infty$ .

(b)  $Y_{N_f} = V V^T F U U^T + U U^T F V V^T = (I - P)F P + P F(I - P)$ . Since  $X$  commutes with  $F$ ,  $[X, F P] = F[X, P]$ ,  $[X, P F] = [X, P]F$  and  $[X, P F P] = [X, P]F P + P F[X, P]$ , which gives the commutator formula. Then  $\|[X, T_f]\|_F \leq 2\|f\|_\infty \|[X, P]\|_F$ , i.e. the first bound (Lemma 2.2). For  $f = e_k$  and  $E_k = \text{Diag}(e_k)$ ,  $E_k[X, P]$  and  $[X, P]E_k$  are the  $k$ th row and column of  $[X, P]$ , each of squared norm  $\sum_j P_{kj}^2 (x_k - x_j)^2$ .  $\square$

### 5.3 The second-order mean and the drift

**Lemma 5.6** (Mean).  $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - b_*$  with  $b_* := \mathcal{A}m_*$ , where

$$b_0 = \frac{1}{n}L(p^{-1}), \quad \|b_0\|_\infty \leq \frac{12\varepsilon}{n}, \quad m_* = \frac{1}{n} \sum_{\mu>0} \frac{r_\mu}{\mu} \Gamma_{f_y}, \quad \|m_*\|_F^2 \leq \frac{2a}{\rho}.$$

*Proof.* For  $G \sim N(0, I)$  on  $\mathcal{Z}$ ,  $\mathbb{E}\|G^T v_i\|^2 = d(1 - p_i)$  and  $\mathbb{E}\|Gu_i\|^2 = (n - d)p_i$ , so the unfiltered covariance  $I/n$  contributes  $a\mathbf{1} - p$ . The covariance  $\Omega = \sum_j \zeta_j \otimes \zeta_j$ ,  $\zeta_j = \bar{\mathcal{A}}^* e_j = v_j u_j^T / \sqrt{p_j}$ , contributes

$$\frac{1}{n} \sum_j q_i(\zeta_j) = \frac{1}{n} \sum_j \frac{(I - P)_{ij}^2 p_j - (1 - p_j) P_{ij}^2}{p_j} = \frac{1}{n} \left( 1 - \sum_j \frac{P_{ij}^2}{p_j} \right) = \frac{1}{n} (Lp^{-1})_i,$$

using  $\sum_j (I - P)_{ij}^2 = 1 - p_i$  and  $\sum_j P_{ij}^2 = p_i$ . Here  $(Lp^{-1})_i = \sum_j P_{ij}^2 (p_i^{-1} - p_j^{-1})$  and  $|p_i^{-1} - p_j^{-1}| \leq 2\varepsilon a / ((1 - \varepsilon)a)^2 \leq 8\varepsilon/a$ , so  $|(Lp^{-1})_i| \leq 12\varepsilon$ . Finally  $R_\rho/n$  contributes  $\frac{1}{n} \sum_\mu r_\mu q(Z_y) = \frac{1}{n} \sum_\mu \frac{r_\mu}{\mu} q(N_{f_y}) = \mathcal{A}m_*$  by Lemma 5.5(a), and with  $\|N_{f_y}\|_F = \sqrt{\mu}$ , (5.3) and Lemma 5.3(c),

$$\|m_*\|_F \leq \frac{2}{n} \sqrt{\frac{2}{a}} \sum_{\mu>0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{2}{n} \sqrt{\frac{2}{a}} \cdot \frac{d}{2\sqrt{\rho}} = \sqrt{\frac{2a}{\rho}}. \quad \square$$

*Remark 5.7* (Why a drift). The filter removes covariance from the directions  $Z_y$ , and with it their share  $t^2 b_* = t^2 \mathcal{A}m_*$  of the second-order drift towards  $a\mathbf{1}$ . This bias need not be below the tolerance  $\sigma/(12\Theta)$  used later: for instance, for many small blocks joined through a common row it is of order  $at^2$  at that row. (Since  $\Gamma_f = V^T[(ff^T) \circ (I - 2P)]U$  is quadratic in  $f$ , and kernel modes  $Sy = 0$  have  $N_{f_y} = 0$  and hence  $\Gamma_{f_y} = 0$ , the spectral identity  $\sum_y \frac{1-\mu}{\rho+\mu} yy^T = (I - S)(\rho I + S)^{-1}$  shows that  $m_*$  does not depend on the choice of eigenbasis:  $m_* = \frac{1}{n} V^T[(D_p^{-1/2}(I - S)(\rho I + S)^{-1} D_p^{-1/2}) \circ (I - 2P)]U$ .) However,  $t^2 b_*$  is exactly the *first-order* diagonal change produced by the deterministic horizontal direction  $\frac{t}{2} m_*$ , whose norm is  $\frac{t}{2} \|m_*\|_F \leq \sqrt{a}/2/D$ . Adding this direction to the noise cancels the bias exactly, at the price of cross terms of size  $O(\sigma/D)$ .

### 5.4 The expected graph

Since  $\mathbb{E}Y_{ij}^2$  is linear in the covariance,  $\mathbb{E}_\Sigma \mathcal{L}(Y) = \sum_k \lambda_k \mathcal{L}(Y_{\psi_k})$  for every representation  $\Sigma = \sum_k \lambda_k \psi_k \otimes \psi_k$  with real  $\lambda_k$ ; we use it for  $C_\rho/n = (I - \Omega - R_\rho)/n$  with  $\Omega = \sum_j \zeta_j \otimes \zeta_j$ .

**Lemma 5.8.**  $\mathbb{E}\mathcal{L}(Y) \succeq \frac{a}{4n} \mathbf{K} - \left( \frac{2}{n} + \frac{8}{d} + \frac{8}{\rho} \right) L$ .

*Proof. Unfiltered noise.* For  $Z_0 \sim N(0, I/n)$  and  $i \neq j$ ,  $(Y_{Z_0})_{ij} = v_i^T Z_0 u_j + v_j^T Z_0 u_i$  has variance  $n^{-1} \|v_i u_j^T + v_j u_i^T\|_F^2 = n^{-1} (p_i + p_j - 2p_i p_j - 2P_{ij}^2)$ . As  $p \leq \frac{3}{4}$ ,  $p_i + p_j - 2p_i p_j \geq \frac{1}{4} (p_i + p_j) \geq \frac{a}{4}$ , so  $\mathbb{E}\mathcal{L}(Y_{Z_0}) \succeq \frac{a}{4n} \mathbf{K} - \frac{2}{n} L$ .

*Base loss.*  $Y_{\zeta_k} = T_{e_k} / \sqrt{p_k}$ , so by Lemma 5.5(b) and  $p_k \geq a/2$ ,  $\sum_k x^T \mathcal{L}(Y_{\zeta_k}) x \leq \frac{2}{a} \cdot 2 \sum_{k,j} P_{kj}^2 (x_k - x_j)^2 = \frac{8}{a} x^T L x$ ; the base part  $\Omega/n$  therefore removes at most  $\frac{8}{d} L$ .

*Residual loss.*  $\mathcal{L}(Y_{Z_y}) = \mu^{-1} \mathcal{L}(T_{f_y}) \preceq \frac{8}{a\mu} L$  by Lemma 5.5(b) and (5.3), so  $R_\rho/n$  removes at most  $\frac{8}{an} \sum_\mu \frac{r_\mu}{\mu} L \leq \frac{8}{\rho} L$ .  $\square$

## 5.5 A good sample

Let  $\ell_i = \frac{1}{n} \sum_{\mu} r_{\mu} \|(Y_{Z_y})_i\|^2$  be the variance that  $R_{\rho}/n$  removes from row  $i$ . Since  $\|Y_{Z_y}\|_{\mathbb{F}}^2 = 2$ ,  $\sum_i \ell_i \leq \frac{2}{n} \sum_{\mu} r_{\mu} \leq 2a$ . Fix  $\delta_0 = 1/6400$  and let

$$\mathfrak{B} = \{i : \ell_i > \delta_0 a\}, \quad b = |\mathfrak{B}| \leq 2/\delta_0 =: b_{\max},$$

a deterministic set depending only on  $U$  and  $\rho$ . Write  $\mathcal{C}(Y) = \sum_{i < j} 2P_{ij}Y_{ij}(e_i - e_j)(e_i - e_j)^T$ , so that  $\mathcal{L}(P + tY) = L + t\mathcal{C}(Y) + t^2\mathcal{L}(Y)$ .

**Lemma 5.9.** *There is an absolute  $h \in (0, 1)$  and, for each  $\eta' \in (0, \frac{1}{32}]$ , a constant  $A_{\eta'}$  such that the following holds for all  $\rho > 0$  and  $0 < t \leq 1$  whenever  $d \geq A_{\eta'} \log(2n)$ . With positive probability:*

- (S1)  $\|Z\|_{\text{op}} \leq 16$ ,  $\max_i \|Z^T v_i\|^2 \leq 3a$  and  $\max_i \|Y_i\|^2 \leq 800a$ ;
- (S2)  $\max_i |(\mathcal{A}Z)_i| \leq 3\sqrt{\rho a \log(2n)/n}$  and  $\max_i |q_i(Z) - \mathbb{E}q_i(Z)| \leq \eta'a$ ;
- (S3) every  $i \notin \mathfrak{B}$  has at least  $0.95n$  indices  $j \neq i$  with  $|P_{ij} + tY_{ij}| \geq ht\sqrt{a/n}$ ;
- (S4) for every  $x$  supported on  $\mathfrak{B}$ ,

$$|tx^T \mathcal{C}(Y)x| \leq \eta'(x^T Lx + at^2\|x\|^2), \quad |x^T (\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y))x| \leq \eta'a\|x\|^2.$$

*Proof.* Since  $C_{\rho} \preceq I$ , every linear functional  $\langle Z, \Psi \rangle$  has variance at most  $\|\Psi\|_{\mathbb{F}}^2/n$ ; we use the tools of [Appendix A](#).

(S1): The net bound ([Lemma A.1\(c\)](#)) with  $\sigma^2 = 1/n$  gives  $\mathbb{P}(\|Z\|_{\text{op}} > 16) \leq 2 \cdot 5^n e^{-8n}$ . The vector  $Z^T v_i$  has covariance  $\preceq n^{-1}I_d$  and trace  $\leq a$ ; [Lemma A.1\(a\)](#) with  $u = 2a$  (so  $\min\{u^2/\|\cdot\|_{\mathbb{F}}^2, u/\|\cdot\|_{\text{op}}\} \geq 2d$ ) and a union bound give  $\max_i \|Z^T v_i\|^2 \leq 3a$  off probability  $2ne^{-cd}$ . Then  $\|Y_i\| \leq \|Z^T v_i\| + \|Zu_i\| \leq \sqrt{3a} + 16\sqrt{3a/2}$ .

(S2):  $(\mathcal{A}Z)_i$  is Gaussian with variance  $\leq \rho p_i/n \leq 3\rho a/(2n)$  ([Lemma 5.3\(b\)](#)); a union bound gives the first claim off probability  $2n(2n)^{-3}$ . The form  $q_i(Z) = \langle Z, \mathbf{M}_i Z \rangle$ ,  $\mathbf{M}_i Z = v_i v_i^T Z - Zu_i u_i^T$ , has  $\|\mathbf{M}_i\|_{\text{op}} \leq 1$  and  $\|\mathbf{M}_i\|_{\mathbb{F}}^2 \leq 2(d + np_i^2) \leq 7d$ ; after the covariance factor  $(C_{\rho}/n)^{1/2}$  these become  $\leq 1/n$  and  $\leq \sqrt{7d}/n$ , and [Lemma A.1\(a\)](#) with  $u = \eta'a$  gives failure probability  $2n \exp(-c\eta'^2 d)$ .

(S3): *Variance bookkeeping.* By the proof of [Lemma 5.8](#), for  $i \neq k$  with  $P_{ik}^2 \leq a/16$  the unfiltered variance of  $Y_{ik}$  is at least  $a/(8n)$ ; at most 24 indices  $k$  have  $P_{ik}^2 > a/16$ . Row  $i$  of  $T_{e_j}$  is  $\delta_{ij}e_j^T P + P_{ij}e_j^T - 2P_{ij}e_j^T P$ , of squared norm at most  $3(\delta_{ij}p_j + P_{ij}^2 + 4P_{ij}^2 p_j)$ , so the base part removes from row  $i$  total variance  $\frac{1}{n} \sum_j \|e_i^T T_{e_j}\|^2/p_j \leq \frac{6}{an} \cdot 5p_i \leq \frac{45}{n}$ , which exceeds  $\frac{a}{32n}$  at no more than  $1440n/d$  positions. For  $i \notin \mathfrak{B}$  the residual part removes at most  $\delta_0 a$  from row  $i$ , hence more than  $\frac{a}{32n}$  at no more than  $32\delta_0 n$  positions. So for  $d \geq 10^6$  (hence  $n \geq 2 \cdot 10^6$ , and the excluded indices number at most  $25 + 0.0065n \leq 0.01n$ ) every  $i \notin \mathfrak{B}$  has at least  $0.99n$  indices  $k \neq i$  with  $\text{Var}(Y_{ik}) \geq \frac{a}{16n}$ .

*Small ball.* Let  $\phi : \mathbb{R} \rightarrow [0, 1]$  be  $C^1$ , equal to 1 on  $[-h, h]$ , vanishing outside  $[-2h, 2h]$ , with  $|\phi'| \leq 2/h$ , and for  $i \notin \mathfrak{B}$  let  $F_i = \frac{1}{n} \sum_{k \neq i} \phi((P_{ik} + tY_{ik})/(t\sqrt{a/n}))$ . By the density bound for a Gaussian of variance  $\geq \frac{at^2}{16n}$ ,  $\mathbb{E}F_i \leq 0.01 + \frac{4h}{\sqrt{2\pi/16}} \leq 0.01 + 7h$ . As a function of the row  $(Y_{ik})_k$ ,  $F_i$  has gradient norm at most  $\frac{1}{n} \cdot \frac{2}{h} \cdot \sqrt{n/a} \cdot \sqrt{n} = 2/(h\sqrt{a})$ ; the row depends on  $Z$  through a linear map of norm  $\leq 2$ , and  $Z = (C_{\rho}/n)^{1/2}g$  with  $g$  standard. So  $F_i$  is  $4/(h\sqrt{d})$ -Lipschitz in  $g$ , and [Lemma A.1\(b\)](#) gives  $\mathbb{P}(F_i > \mathbb{E}F_i + 0.01) \leq 2 \exp(-ch^2 d)$ . Take  $h = 1/500$ , so that  $\mathbb{E}F_i + 0.01 \leq 0.04$ , and a union bound over  $i$ . On the complement, at most  $0.04n$  indices  $k \neq i$  have  $|P_{ik} + tY_{ik}| < ht\sqrt{a/n}$ , so at least  $n - 1 - 0.04n \geq 0.95n$  have the reverse inequality.

(S4): Only second moments are needed, because  $\mathfrak{B}$  is deterministic and bounded. Let  $\Pi_{\mathfrak{B}}$  be the coordinate projection onto  $\mathbb{R}^{\mathfrak{B}}$ ,  $J_{\mathfrak{B}} = L_{\mathfrak{B}\mathfrak{B}} + at^2 I_{\mathfrak{B}}$  and  $\omega_{ij} = J_{\mathfrak{B}}^{-1/2} \Pi_{\mathfrak{B}}(e_i - e_j)$ , so  $\|\omega_{ij}\|^2 \leq 2/(at^2)$  and  $\sum_{i < j} P_{ij}^2 \omega_{ij} \omega_{ij}^T = J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2} \preceq I$ . For  $x$  supported on  $\mathfrak{B}$ ,  $tx^T \mathcal{C}(Y)x = \langle J_{\mathfrak{B}}^{1/2} x, \Xi J_{\mathfrak{B}}^{1/2} x \rangle$  with  $\Xi = \sum_{i < j} 2t P_{ij} Y_{ij} \omega_{ij} \omega_{ij}^T$ , and  $x^T J_{\mathfrak{B}} x = x^T L x + at^2 \|x\|^2$ . The coefficients  $(Y_{ij})_{i < j}$  have covariance  $\preceq \frac{2}{n} I$  (the map  $Z \mapsto Y_Z$  is  $\sqrt{2}$  times an isometry), so

$$\mathbb{E} \|\Xi\|_{\text{F}}^2 \leq \frac{2t^2}{n} \sum_{i < j} 4P_{ij}^2 \|\omega_{ij}\|^4 \leq \frac{16}{an} \text{tr}(J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2}) \leq \frac{16b}{d},$$

and Markov's inequality gives the first half off probability  $16b/(d\eta^2)$ . The principal  $\mathfrak{B}$ -block of  $\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)$  has diagonal entries  $\sum_{j \neq i} (Y_{ij}^2 - \mathbb{E}Y_{ij}^2)$ , a centred Gaussian quadratic form of variance at most  $2 \cdot \frac{2}{n} \cdot \mathbb{E}\|Y_i\|^2 \leq \frac{12a}{n}$  (Lemma A.1(a), with  $\mathbb{E}\|Y_i\|^2 \leq a(1-p_i) + (1-a)p_i \leq 3a$ ), and off-diagonal entries  $-(Y_{ik}^2 - \mathbb{E}Y_{ik}^2)$  of variance at most  $8/n^2$ . Its expected squared Frobenius norm is at most  $20b^2 a/n = 20b^2 a^2/d$ , and Markov gives the second half off probability  $20b^2/(d\eta^2)$ .

Finally choose  $A_{\eta'}$  so large that  $d \geq A_{\eta'} \log(2n)$  implies  $d \geq 10^6$  and all failure probabilities above are at most  $\frac{1}{10}$ ; their sum is less than 1.  $\square$

## 5.6 The drifted retraction

Fix  $D \geq 12$  and  $t \in (0, 1]$ , set  $\rho = D^2 t^2$ , take a sample  $Z$  as in Lemma 5.9, and let  $m_*$  be as in Lemma 5.6. Define

$$W = Z + \frac{t}{2} m_*, \quad G = I + t^2 W^T W, \quad U_t = (U + tVW)G^{-1/2}, \quad P_t = U_t U_t^T.$$

Since  $U^T V = 0$ ,  $(U + tVW)^T (U + tVW) = G$  and  $U_t$  is Parseval.

**Lemma 5.10.** *There is an absolute  $C_E$  (depending only on the bounds in (S1) and on  $D \geq 12$ ) such that:*

- (a)  $\|U_t - U\|_{\text{F}}^2 \leq C_E t^2 d$ ;
- (b) every row of  $\mathbf{E} = P_t - P - tY$  satisfies  $\sum_j \mathbf{E}_{ij}^2 \leq at^2(D^{-2} + C_E t^2)$ ;
- (c) with  $\|r\|_{\infty} \leq C_E at^3$ ,

$$\text{diag } P_t - a\mathbf{1} = (1 - t^2)(p - a\mathbf{1}) + 2tAZ - t^2 b_0 + t^2(q(Z) - \mathbb{E}q(Z)) + t^3 q(Z, m_*) + \frac{t^4}{4} q(m_*) + r;$$

$$(d) \|\text{diag } P_t - a\mathbf{1}\|_{\infty} \leq at^2 \left( \frac{\varepsilon}{t^2} + 6D\sqrt{\log(2n)/d} + \frac{12\varepsilon}{d} + \eta' + \frac{40}{D} + \frac{1}{2D^2} + C_E t \right).$$

*Proof.* By Lemma 5.6,  $\frac{t}{2} \|m_*\|_{\text{op}} \leq \frac{t}{2} \|m_*\|_{\text{F}} \leq \sqrt{a/2}/D \leq \frac{1}{24}$ , so with (S1):  $\|W\|_{\text{op}} \leq 17$ ,  $\|W\|_{\text{F}} \leq 17\sqrt{d}$ ,  $\|W^T v_i\| \leq 2\sqrt{a}$ ,  $\|W u_i\| \leq 17\sqrt{3a/2}$ , and  $\|(Y_{m_*})_i\|^2 \leq \|m_*\|_{\text{op}}^2 \leq 2a/\rho$ . Hence  $\|G^{-1} - I\|_{\text{op}}$  and  $\|G^{-1/2} - I\|_{\text{op}}$  are at most  $289t^2$ ,  $\|U + tVW\|_{\text{op}} \leq 18$  and  $\|u_i + tW^T v_i\| \leq 4\sqrt{a}$ .

$$(a) \|U_t - U\|_{\text{F}} \leq \|U + tVW\|_{\text{F}} \|G^{-1/2} - I\|_{\text{op}} + t\|W\|_{\text{F}} \leq Ct\sqrt{d}.$$

(b)  $\mathbf{E} = \frac{t^2}{2} Y_{m_*} + \mathbf{E}'$  with  $\mathbf{E}' = P_t - P - tY_W = (U + tVW)(G^{-1} - I)(U + tVW)^T + t^2 VWW^T V^T$ , whose rows have norm at most  $Ct^2 \sqrt{a}$ . The rows of  $\frac{t^2}{2} Y_{m_*}$  have squared norm at most  $\frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$ . Use  $(\alpha + \beta)^2 \leq 2\alpha^2 + 2\beta^2$ .

(c) Let  $\xi_i = u_i + tW^T v_i$ , the  $i$ th row of  $U + tVW$ . Inserting  $G^{-1} = I - t^2 W^T W + t^4 (W^T W)^2 G^{-1}$  into  $(P_t)_{ii} = \xi_i^T G^{-1} \xi_i$  and using  $u_i^T W^T v_i = (\mathcal{A}W)_i$ ,

$$(P_t)_{ii} = p_i + 2t(\mathcal{A}W)_i + t^2 q_i(W) - 2t^3 \langle W u_i, W W^T v_i \rangle - t^4 \|W W^T v_i\|^2 + t^4 \xi_i^T (W^T W)^2 G^{-1} \xi_i,$$

and the last three terms are  $O(at^3)$ . Now  $2tAW = 2tAZ + t^2Am_*$ ,  $q(W) = q(Z) + tq(Z, m_*) + \frac{t^2}{4}q(m_*)$ , and  $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - Am_*$  by Lemma 5.6; the terms  $\pm t^2Am_*$  cancel.

(d) In (c):  $(1 - t^2)\|p - a\mathbf{1}\|_\infty \leq \varepsilon a$ ;  $2t\|AZ\|_\infty \leq 6Dt\sqrt{a\log(2n)/n}$  by (S2), which is  $6Dat^2\sqrt{\log(2n)/d}$ ;  $t^2\|b_0\|_\infty \leq 12\varepsilon t^2/n$ ; the fluctuation is at most  $\eta'at^2$ ; by (5.4) and (S1),  $t^3|q_i(Z, m_*)| \leq t^3\sqrt{800a}\sqrt{2a/\rho} = 40at^2/D$  and  $\frac{t^4}{4}|q_i(m_*)| \leq \frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$ .  $\square$

**Lemma 5.11** (Graph of the seed). *There are absolute  $t_* \in (0, 1]$  and  $C_{\text{core}}$  such that the following holds. Let  $\eta' \leq \frac{1}{32}$ ,  $d \geq A_{\eta'}\log(2n)$ ,  $D \geq 13/h$ ,  $b \leq n/10$  and  $t \leq t_*$ , and let  $Z$  satisfy (S1)–(S4). Then  $H(L_{P_t}) \leq C_{\text{core}}/(at^2)$ .*

*Proof. Core.* In a row  $i \notin \mathfrak{B}$ , by Lemma 5.10(b), at most  $at^2(D^{-2} + C_E t^2)/(\frac{h^2}{4}\frac{at^2}{n}) \leq 0.05n$  indices have  $|\mathbf{E}_{ij}| \geq \frac{h}{2}t\sqrt{a/n}$ , once  $D \geq 13/h$  and  $t^2 \leq h^2/(160C_E)$ . With (S3), every  $i \notin \mathfrak{B}$  has at least  $0.9n$  indices  $j \neq i$  with  $(P_t)_{ij}^2 \geq \frac{h^2}{4}\frac{at^2}{n}$ .

*Block.* For  $x$  supported on  $\mathfrak{B}$ , Lemma 5.8,  $x^TKx \geq (n - b)\|x\|^2$  and (S4) give

$$x^T\mathcal{L}(P + tY)x \geq \left(1 - \eta' - t^2\left(\frac{2}{n} + \frac{8}{d}\right) - \frac{8}{D^2}\right)x^TLx + at^2\left(\frac{1}{4}\left(1 - \frac{b}{n}\right) - 2\eta'\right)\|x\|^2 \geq \frac{at^2}{8}\|x\|^2,$$

since the first coefficient is positive and  $\frac{1}{4} \cdot \frac{9}{10} - \frac{1}{16} \geq \frac{1}{8}$ . By (5.1) and Lemma 5.10(b),  $L_{P_t} = \mathcal{L}(P + tY + \mathbf{E}) \succeq \frac{1}{2}\mathcal{L}(P + tY) - 2at^2(D^{-2} + C_E t^2)I$ , which is at least  $\frac{at^2}{32}$  on vectors supported on  $\mathfrak{B}$  once  $t^2 \leq 1/(128C_E)$ .

*Barrier.* Lemma 2.10 with  $\mathcal{B} = \mathfrak{B}$ , part (i) with  $\vartheta = 0.4$ ,  $\gamma = h^2at^2/4$ , and part (ii) with  $\lambda = at^2/32$ , gives  $H(L_{P_t}) \leq \frac{1+b}{0.1h^2at^2} + \frac{32\sqrt{b}}{at^2}$ . Put  $C_{\text{core}} = 10(1 + b_{\max})/h^2 + 32\sqrt{b_{\max}}$ .  $\square$

## 5.7 Proof of Theorem 5.1

*Constants.* The numbers  $\delta_0, b_{\max}, h, C_E, t_*, C_{\text{core}}$  are absolute and do not depend on  $D$  (for  $D \geq 12$ ). Put  $\Theta = \max\{2, C_{\text{core}}, C_E\}$  and choose, in this order:  $D \geq \max\{13/h, 500\Theta\}$ , so that  $\frac{40}{D} + \frac{1}{2D^2} \leq \frac{1}{12\Theta}$ ;  $K_0 = 12\Theta$  and  $\eta' = \min\{\frac{1}{32}, \frac{1}{12\Theta}\}$ ;  $A \geq A_{\eta'}$  so large that  $d \geq A\log(2n)$  implies  $6D\sqrt{\log(2n)/d} \leq \frac{1}{12\Theta}$  and  $b_{\max} \leq n/10$ ; and  $\varepsilon_0 \leq \frac{1}{2}$  so that  $t = \sqrt{K_0\varepsilon}$  satisfies  $t \leq t_*$ ,  $C_E t \leq \frac{1}{12\Theta}$  and  $12\varepsilon \leq \frac{1}{12\Theta}$  for  $\varepsilon \leq \varepsilon_0$ . None of these choices depends on  $n, d, \varepsilon$  or  $U$ .

*Seed.* With  $t^2 = K_0\varepsilon$ , Lemma 5.10(d) bounds the diagonal error of  $P_t$  by  $at^2 \cdot \frac{6}{12\Theta} = \frac{at^2}{2\Theta}$ ; Lemma 5.11 bounds the barrier constant by  $\Theta/(at^2)$ ; and Lemma 5.10(a) bounds  $\|U_t - U\|_{\text{F}}^2$  by  $\Theta t^2 d$ . So  $U_t$  is a  $(\Theta, t)$ -seed for  $U$ , and Corollary 2.13 gives an ENP frame  $\hat{U}$  with  $\|\hat{U} - U\|_{\text{F}}^2 \leq 3\Theta t^2 d = 36\Theta^2 \varepsilon d$ .  $\square$

*Remark 5.12.* The drift keeps the spectral input local. The bias  $t^2b_*$  could also be removed after the perturbation, for instance by a second scaling; since  $\langle x, b_* \rangle^2 \leq \frac{2a}{\rho}x^TLx$  this is possible, but it requires two-sided control of  $L_{P_t}$  on all of  $\mathbf{1}^\perp$ . With the drift, the only spectral input is the gap on the bounded deterministic block  $\mathfrak{B}$ , which needs second moments only.

## 6 Assembly

We use the two seeds through the following equal-norm formulation.

**Proposition 6.1.** *There is an absolute constant  $C_0$  such that every  $X \in \mathbb{R}^{n \times d}$  with  $1 \leq d$ ,  $2d \leq n$  and  $\|x_i\|^2 = a$  for all  $i$  admits an ENP frame  $W$  with*

$$\|X - W\|_{\text{F}}^2 \leq C_0 d \|X^T X - I\|_{\text{op}}.$$

*Proof.* Let  $\eta = \|X^T X - I\|_{\text{op}}$ ; if  $\eta = 0$  take  $W = X$ . Let  $A, \varepsilon_0, C_m$  be the constants of [Theorem 5.1](#) and  $B, c_*, C_*$  those of [Theorem 4.1](#), and fix an integer  $d_0 \geq 2$  with  $A \log(2Bd^2) \leq d$  for all  $d \geq d_0$ .

*Case  $d < d_0$ .* [Lemma 3.1](#) gives  $\|X - W\|_{\text{F}}^2 \leq \eta d(d-1) \leq d_0 \eta d$ .

*Case  $d \geq d_0$ ,  $n \geq Bd^2$ .* If  $\eta \leq c_*$  use [Theorem 4.1](#); otherwise [Corollary 3.2](#) gives cost  $4d \leq (4/c_*)\eta d$ .

*Case  $d \geq d_0$ ,  $n < Bd^2$ .* Then  $A \log(2n) \leq d$ . If  $\eta \leq \varepsilon_0/4$ , let  $U = X(X^T X)^{-1/2}$ . The singular values of  $X$  lie in  $[\sqrt{1-\eta}, \sqrt{1+\eta}]$ , so  $\|X - U\|_{\text{F}}^2 \leq d\eta^2$ , and  $\|u_i\|^2 = x_i^T (X^T X)^{-1} x_i \in [a/(1+\eta), a/(1-\eta)]$ , so  $|\|u_i\|^2 - a| \leq 2\eta a$ . [Theorem 5.1](#) with  $\varepsilon = 2\eta$  gives an ENP  $W$  with  $\|U - W\|_{\text{F}}^2 \leq 2C_m \eta d$ , and  $\|X - W\|_{\text{F}}^2 \leq 2d\eta^2 + 4C_m \eta d$ . If  $\eta > \varepsilon_0/4$  use the trivial cost  $4d \leq (16/\varepsilon_0)\eta d$ . Altogether  $C_0 = \max\{d_0, C_*, 4/c_*, 2 + 4C_m, 16/\varepsilon_0\}$  works.  $\square$

*Proof of [Theorem 1.2](#).* Replacing  $(P, Q)$  by  $(I - P, I - Q)$  preserves  $\beta$ , ranks of complements and distances, so we may assume  $d \leq n/2$ ; the case  $d = 0$  is trivial. Write  $P = UU^T$  with  $U^T U = I$ ,  $p_i = \|u_i\|^2$  and  $\delta = \beta/a$ . If  $\delta \geq \frac{1}{2}$ , any ENP  $W$  ([Corollary 3.2](#)) gives  $\|P - WW^T\|_{\text{F}}^2 \leq 2d \leq 4n\beta$ . Otherwise let  $x_i = \sqrt{a} u_i / \|u_i\|$ . Then  $\|X - U\|_{\text{F}}^2 = \sum_i (\sqrt{p_i} - \sqrt{a})^2 \leq \sum_i (p_i - a)^2 / a \leq \delta^2 d$  and  $X^T X = U^T \text{Diag}(a/p) U$ , so  $\|X^T X - I\|_{\text{op}} \leq \max_i |a/p_i - 1| \leq \delta/(1-\delta) \leq 2\delta$ . [Proposition 6.1](#) gives an ENP  $W$  with  $\|U - W\|_{\text{F}}^2 \leq 2\delta^2 d + 4C_0 \delta d \leq (1 + 4C_0)n\beta$ , using  $\delta d = n\beta$ . Take  $Q = WW^T$ ; by [Lemma 2.1](#),  $\|P - Q\|_{\text{F}}^2 \leq 2(1 + 4C_0)n\beta$ . So [Theorem 1.2](#) holds with  $C_P = \max\{4, 2 + 8C_0\}$ .  $\square$

*Proof of [Theorem 1.1](#).* If  $\varepsilon \geq \frac{1}{2}$ , an ENP  $W$  gives  $\|X - W\|_{\text{F}}^2 \leq 2\|X\|_{\text{F}}^2 + 2d \leq 6d \leq 12\varepsilon d$ , since  $\|X\|_{\text{F}}^2 = \text{tr } X^T X \leq (1 + \varepsilon)d$ . Let  $\varepsilon < \frac{1}{2}$  and  $U = X(X^T X)^{-1/2}$ ,  $P = UU^T$ . As above  $\|X - U\|_{\text{F}}^2 \leq \varepsilon^2 d$  and  $P_{ii} = x_i^T (X^T X)^{-1} x_i \in [\frac{1-\varepsilon}{1+\varepsilon}a, \frac{1+\varepsilon}{1-\varepsilon}a]$ , so  $\beta = \max_i |P_{ii} - a| \leq 4\varepsilon a$ . [Theorem 1.2](#) gives  $Q$  with  $\|P - Q\|_{\text{F}}^2 \leq 4C_P \varepsilon d$ , and by [Lemma 2.1](#) there is an isometry  $W$  spanning  $Q$ —hence ENP—with  $\|U - W\|_{\text{F}}^2 \leq \|P - Q\|_{\text{F}}^2$ . Then  $\|X - W\|_{\text{F}}^2 \leq 2\varepsilon^2 d + 8C_P \varepsilon d$ , so [Theorem 1.1](#) holds with  $C = \max\{12, 1 + 8C_P\}$ .  $\square$

## A Gaussian tools

**Lemma A.1.** *The following hold with absolute constants.*

(a) (Quadratic forms.) For  $g$  standard Gaussian and  $\mathbf{A}$  symmetric,  $\text{Var}(g^T \mathbf{A} g) = 2\|\mathbf{A}\|_{\text{F}}^2$  and

$$\mathbb{P}\{|g^T \mathbf{A} g - \text{tr } \mathbf{A}| > u\} \leq 2 \exp\left[-\frac{1}{8} \min\left\{\frac{u^2}{\|\mathbf{A}\|_{\text{F}}^2}, \frac{u}{\|\mathbf{A}\|_{\text{op}}}\right\}\right].$$

For a Gaussian vector  $\zeta = \Sigma^{1/2} g$ , apply this to  $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ , whose norms are at most  $\|\Sigma\|_{\text{op}}$  times those of  $\mathbf{A}$ .

(b) (Lipschitz functions.) If  $F$  is  $C^1$  with  $\|\nabla F\| \leq \ell$  and  $g$  is standard Gaussian, then  $\mathbb{P}\{|F(g) - \mathbb{E}F(g)| > u\} \leq 2 \exp(-2u^2/(\pi^2 \ell^2))$ .

(c) (Nets.) If  $Z$  is an  $r \times s$  centred Gaussian matrix with  $\text{Cov}(\text{vec } Z) \preceq \sigma^2 I$ , then  $\mathbb{P}\{\|Z\|_{\text{op}} > u\} \leq 2 \cdot 5^{r+s} \exp(-u^2/(32\sigma^2))$ .

*Proof.* (a) Diagonalise  $\mathbf{A}$  with eigenvalues  $\lambda_j$ . For  $|s| \leq 1/(4\|\mathbf{A}\|_{\text{op}})$ ,  $\log \mathbb{E} e^{s(g^T \mathbf{A} g - \text{tr } \mathbf{A})} = \sum_j [-s\lambda_j - \frac{1}{2} \log(1 - 2s\lambda_j)] \leq 2s^2 \|\mathbf{A}\|_{\text{F}}^2$ , since  $-x - \frac{1}{2} \log(1 - 2x) = \sum_{k \geq 2} (2x)^k / (2k) \leq 2x^2$  for  $|x| \leq \frac{1}{4}$ . The exponential Markov inequality with  $s = \min\{u/(4\|\mathbf{A}\|_{\text{F}}^2), 1/(4\|\mathbf{A}\|_{\text{op}})\}$ , applied to both signs, gives the tail. The variance follows from  $\mathbb{E} g_j^4 = 3$ .

(b) Let  $g'$  be an independent copy,  $g_\theta = g \cos \theta + g' \sin \theta$  and  $\dot{g}_\theta = -g \sin \theta + g' \cos \theta$ ; for each  $\theta$  the pair  $(g_\theta, \dot{g}_\theta)$  is standard Gaussian. By the fundamental theorem of calculus and Jensen on  $[0, \pi/2]$ ,

$$\mathbb{E} e^{s(F(g') - F(g))} \leq \frac{2}{\pi} \int_0^{\pi/2} \mathbb{E} \exp \left[ \frac{\pi s}{2} \langle \nabla F(g_\theta), \dot{g}_\theta \rangle \right] d\theta \leq e^{\pi^2 s^2 \ell^2 / 8}.$$

Jensen in  $g'$  gives  $\mathbb{E} e^{s(F(g) - \mathbb{E} F)} \leq e^{\pi^2 s^2 \ell^2 / 8}$ ; optimise in  $s$ . All expectations are finite since  $|F(x)| \leq |F(0)| + \ell \|x\|$ , and Fubini applies to the  $\theta$ -integral.

(c) Let  $\mathcal{N}_1, \mathcal{N}_2$  be  $\frac{1}{2}$ -nets of the unit spheres of  $\mathbb{R}^r$  and  $\mathbb{R}^s$  with at most  $5^r, 5^s$  points. For a vector  $w$ , a net point  $y'$  within  $\frac{1}{2}$  of  $w/\|w\|$  has  $\langle w, y' \rangle \geq \frac{1}{2}\|w\|$ ; applying this twice gives  $\|Z\|_{\text{op}} \leq 4 \max_{x \in \mathcal{N}_1, y \in \mathcal{N}_2} |x^T Z y|$ . Each  $x^T Z y$  is Gaussian with variance  $\leq \sigma^2$ ; a union bound finishes.  $\square$

**Covariance domination.** If a Gaussian vector  $\xi$  has covariance  $\Gamma \preceq \gamma I$  and  $(M_\alpha)$  are matrices, then  $\mathbb{E} \|\sum_\alpha \xi_\alpha M_\alpha\|_F^2 = \text{tr}(\Gamma G) \leq \gamma \sum_\alpha \|M_\alpha\|_F^2$ , where  $G \succeq 0$  is the Gram matrix of the  $M_\alpha$ . Correlated Gaussians enter only as  $\Sigma^{1/2} g$ : in quadratic forms through the last sentence of (a) (Lemma 4.3, Lemma 5.9), in Lipschitz functions through (b), in operator norms through (c), and in Frobenius norms of linear combinations through this identity (Lemma 5.9(S4)).

## B Formalisation

Theorem 1.1 is formalised in Lean 4 with Mathlib; the development accompanies this paper (directory `lean/`). The formal statement is

```
def SharpPaulsenBound : Prop :=
  ∃ C : ℝ, 0 < C ∧
    ∀ (n d : ℕ), 0 < d → d ≤ n →
      ∀ (ε : ℝ) (U : Frame n d), 0 < ε → ε < 1 →
        IsNearlyEqualNormParseval ε U →
          ∃ W : Frame n d, IsEqualNormParseval W ∧
            sqDistance U W ≤ C * ε * (d : ℝ)
```

where `Frame n d` is the type of real  $n \times d$  matrices, Parsevalness is  $U^T U = 1$ , the near-Parseval condition is stated through the quadratic form  $x \mapsto \|Ux\|^2$ , the row-norm conditions refer to  $d/n$  as a real quotient, and `sqDistance` is the unnormalised squared Frobenius distance. The projection form (Theorem 1.2) is derived for arbitrary real symmetric idempotent matrices of rank  $d$ . Both statements are also restated in plain Mathlib vocabulary (positive semidefiniteness, dot products, traces, `Matrix.rank`), and the restatements are derived from them.

**Structure.** The development has about 29,600 lines. A library of about 26,000 lines contains frames, projections and alignment, the squared-Gram Laplacian, diagonal scaling and the log-determinant potential, complementation and normalisation, the Hamilton–Moitra argument (Section 3), Gaussian tools (quadratic forms, Lipschitz functions, nets, covariance calculus), the constrained and row-independent noises and the row-normalisation identity of Section 4, and the filtered covariance, the mean formula and the sampling estimates of Section 5. About 3,600 lines (directory `Paulsen/Linear`) implement the argument specific to this paper:

| paper                                    | Lean (Paulsen/Linear/)                                  |
|------------------------------------------|---------------------------------------------------------|
| Definition 2.6 and Lemma 2.7             | Barrier: HasBarrierBound, barrier_subset_bound          |
| Theorem 2.8                              | Balancing: balancing_cost_barrier                       |
| Lemma 2.10 (main part, (i), (iii))       | Core, ManyRowBarrier                                    |
| Lemma 2.10(ii)                           | BlockBarrier: block_barrier                             |
| Definition 2.12 and Corollary 2.13       | Seed: IsSeed.hasCorrection                              |
| Lemma 4.2                                | ManyRowMoments                                          |
| Lemma 4.4                                | ManyRowRemainder: tangentRemainder_centred_bound        |
| Theorem 4.1                              | ManyRowGraph, ManyRowSample, ManyRowParameters, ManyRow |
| Lemma 5.5(a), Lemma 5.6 ( $m_*$ ), (5.4) | DriftAlgebra: driftGamma, driftMean                     |
| Lemma 5.10                               | DriftRetraction, DriftDiagonal                          |
| Lemma 5.11, Theorem 5.1                  | ModerateBarrier, ModerateDrift                          |
| Proposition 6.1, Theorems 1.1 and 1.2    | Assembly, Main                                          |

**Differences from the text.** The formal proof follows the argument above, with these differences of detail.

- (a) All constants are much larger (for instance  $B = 10^{40}$ ,  $c_* = 10^{-60}$  in Theorem 4.1;  $A = 10^{130}$ ,  $\varepsilon_0 = 10^{-130}$ ,  $D = 10^{40}$  in Theorem 5.1); the balancing cost is  $8\|q-p\|_1$  rather than  $\frac{15}{4}\|q-p\|_1$ ; the net bound is used with  $\|Z\|_{\text{op}} \leq 128$ ; and Lemma 2.10(ii) is used with  $R = b/\lambda$ .
- (b) In Lemma 5.9, the block estimate (S4) is not proved from second moments. Instead the formalisation uses two global events that are also available from the library when  $d \geq A \log 2n$ : the concentration  $\|\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)\|_{\text{op}} \leq a/32$  of the squared Gaussian graph, and the bound  $|tx^T \mathcal{C}(Y)x| \leq \frac{1}{32}(x^T Lx + at^2\|x\|^2)$  for all  $x$ . Together with Lemma 5.8 they give a gap  $\frac{at^2}{32}$  for  $L_{P_t}$  on  $\mathbf{1}^\perp$ , and subtracting the mean gives the gap on vectors supported on  $\mathfrak{B}$  that Lemma 5.11 needs. The rest of the moderate-row argument is as in the text; in particular, nothing is corrected after the drifted retraction.
- (c) In Lemma 4.2 the fourth moment of a centred Gaussian quadratic form is bounded through its moment generating function, which gives a worse constant. One functional,  $a \sum_i (r_i^2 - 1)^2$ , controls both  $\|\Gamma Z\|_{\text{F}}^2$  and  $\|\Lambda X\|_{\text{F}}^2$ .
- (d) Lemma 4.6 is used in a slightly different form: the dense reference graph has  $\frac{17}{20}n$  heavy neighbours at every row, which holds for every  $d \geq 1$ .

**Checks.** The project builds with Lean 4.32.0 and Mathlib v4.32.0. `Audit.lean` checks that the final theorems have the target types; that every theorem of the project depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`; and that the proof of the main theorem uses the barrier-constant toolbox, the centred remainder, the drift and the three-case assembly of Proposition 6.1, and none of the library’s balancing lemmas formulated with Poisson constants. Every module was also replayed in the kernel with `leanchecker`.

## Contributions and acknowledgements

K. Hänni and H. Eberhard proposed the overall strategy: perturb by a random tangent of size  $\sqrt{\varepsilon}$  to create a well-connected squared-Gram graph, and then balance by a diagonal scaling controlled in  $\ell_\infty$ . Astra developed a complete proof along these lines, including both seeds and the reduction to them, together with the Lean library on which the formalisation rests. Claude contributed

the cancelling drift in the moderate-row seed, the centred remainder estimate that makes the many-row seed work at every error level, and the barrier-constant formulation of balancing; wrote this text; and completed the formalisation of the proof given here.

## References

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